

Short View Summary

- **DEFINITION**

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1. • Infectious arthritis is an infection of one or more joints that can be caused by bacteria, viruses, fungi, and parasites.

- **CLINICAL CATEGORIES**

- **Acute Bacterial Arthritis**

- • Overall incidence in native joints is 2 to 21 per 100,000 per year, increased in rheumatoid arthritis (RA) and the elderly. Joint infections in persons who use IV drugs began to increase during the opioid epidemic beginning in 2010 and IV drug use is now the predominate risk factor for septic arthritis.
- • **Infection is usually acquired from hematogenous dissemination, less commonly from direct joint inoculation or from a contiguous focus. Most infections are monoarticular; 10% to 20% are polyarticular. The knee is most often involved.**
- • **Microbiology**
 1. • **Gram-positive organisms, including *Staphylococcus aureus* and *Streptococcus* spp., predominate.**
 2. • **Gram-negative bacilli, in 5% to 20%, are mainly in neonates, the elderly, IV drug users, and the immunocompromised.**
 3. • **Prevalence rates of gonococcal arthritis have decreased. There are two syndromes of joint involvement: monoarticular arthritis and disseminated gonorrhea with febrile tenosynovitis and skin lesions.**
 4. • Lyme disease, especially in settings of large joint effusion with little pain, should be

considered in cases of mono- and oligoarticular arthritis in endemic regions (see **Chapter 247**).

- **Viral Arthritis**

- • It may occur sporadically or in community outbreaks that may be small (e.g., parvovirus B19) or explosive (e.g., chikungunya virus).
- • It is caused by infection of the joint space and/or by immune-mediated inflammation.
- • It is usually an acute symmetrical polyarticular arthritis that occurs simultaneously with other systemic symptoms of viral infection.
- • Causes include parvovirus B19, chikungunya virus, SARS-CoV-2, rubella (which may follow immunization), hepatitis B, hepatitis C, and other viruses, including Ross River virus and O'nyong-nyong virus.
- • Parvovirus B19 and chikungunya virus, in particular, may cause prolonged or chronic arthritis after acute infection.
- • Reactive arthritis has rarely been reported following COVID-19 infection or vaccination.

- **Chronic Infectious Arthritis**

- • It is usually monoarticular, involving the large peripheral joints.
- • It is relatively uncommon compared with acute bacterial arthritis but is increasingly seen in immunocompromised or chronically ill hosts. Etiologies include *Mycobacterium tuberculosis* and nontuberculous mycobacteria. Fungal causes include *Candida*, dimorphic fungi (e.g., *Blastomyces*, *Coccidioides*, and *Sporothrix*), rarely *Cryptococcus*, and molds, among others.

- **CLINICAL EVALUATION**

- • Bacterial arthritis is an emergency and should be considered in any patient with acute monoarticular

arthritis and in bacteremic patients with oligoarticular joint inflammation.

- • The most useful demographic risk factors are presence or absence of recent joint surgery, RA, advanced age, IV drug use, concomitant skin infection, or diabetes mellitus.
- • The most helpful laboratory parameters are the synovial fluid leukocyte count and differential and examination for crystals to evaluate for gout or pseudogout.
- • In patients with acute monoarticular arthritis an elevated serum procalcitonin can be predictive of bacterial joint infection.
- • Diagnostic imaging should include plain film radiography. For prolonged or atypical symptoms or for deep joints, computed tomography or magnetic resonance imaging is recommended.
- • Differential diagnosis includes crystalline, reactive, or traumatic arthritis, in addition to other monoarticular rheumatologic disorders.
- • Identification of the infecting organism in synovial fluid may be made with Gram stain and culture and by multiplex polymerase chain reaction (PCR).
- • For gonococcal arthritis, testing should include nucleic acid amplification testing of pharyngeal, rectal, cervical, or vaginal, and male urethral mucosa or urine.
- THERAPY
 - Acute Bacterial Arthritis
 - • Joint drainage by repeat aspiration, arthroscopy, or arthrotomy is required.
 - • Empirical antibiotic selection should be based on synovial fluid Gram stain and/or

molecular testing (see Table 105.5) or multiplex PCR results.

- • IV therapy, in some instances with an early oral transition, is usually prescribed for a total of 2 to 3 weeks.
- Chronic Arthritis
 - • Therapy for *M. tuberculosis* arthritis is similar to that for pulmonary tuberculosis.
 - • Many fungal arthritides are difficult to treat and result in chronic joint disability. Treatment varies by infecting organism. Experience using the azoles (including new triazoles) and the echinocandins is growing, and in most cases these drugs are replacing amphotericin.
 - • Except for *Cryptococcus* and perhaps *Coccidioides* or *Sporothrix*, fungal arthritis requires surgical drainage and debridement.
- SEPTIC BURSITIS
 - • It predominantly involves the prepatellar, olecranon, and trochanteric bursa.
 - • It can be caused by direct inoculation, by contiguous infection, or, less commonly, from a hematogenous source.
 - • More than 80% of cases are due to *S. aureus*.
 - • Diagnosis is made from the clinical presentation and by aspirating the bursa for fluid analysis, Gram stain, and culture.
 - • Treatment includes antibiotics (usually 14 days) and daily aspiration of the bursa until sterile fluid is obtained. Surgery is occasionally required. Except for severe cases, oral antistaphylococcal agents that have activity against community-acquired methicillin-resistant *S. aureus* are

recommended, pending culture and sensitivity. For moderate-to-severe cases or patients with immunosuppression, IV antibiotics should be selected.

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Infectious arthritis of single or multiple joints may be caused by any of a number of diverse microorganisms. Bacterial arthritis, also known as *suppurative*, *pyogenic*, or *septic arthritis*, is the most common and arguably most important joint infection and is considered a rheumatologic emergency because of its potential for rapid joint destruction with irreversible loss of function. Viral arthritis often involves multiple joints as a component of a systemic infection, and although usually self-limited, it may rarely lead to a chronic polyarticular arthritis. In contrast to the acute occurrence of bacterial and viral arthritis, joint infection caused by mycobacteria and non-*Candida* fungi usually occurs as chronic, slowly progressive monoarticular arthritis. Reactive arthritis (formerly Reiter syndrome), a sterile inflammatory spondyloarthropathy, is occasionally associated with systemic or local infection at a site remote to the joint rather than from direct infection of joint tissue. The clinical manifestations, severity, treatment, and prognosis of septic arthritis are dependent on the identity and virulence of the infecting organism, source of joint infection, and certain underlying host factors, such as immune status, comorbid illness, and abnormal joint architecture from disease or surgery. For patients with infectious arthritis, a prompt and thorough clinical evaluation and early institution of specific treatment are essential to limit long-term sequelae.

This chapter will primarily discuss infections of native joints. For a detailed discussion of infections involving prosthetic arthroplasty see **Chapter 107**. Lyme arthritis caused by disseminated *Borrelia burgdorferi* (Lyme disease) infection is covered in **Chapter 247**.

Acute Bacterial Arthritis

Epidemiology

The reported incidence of native joint septic arthritis in the general population of non–resource-limited regions is between 7 and 21 cases per 100,000 per year.^{1–5} Rates are substantially increasing,

particularly in women and the elderly, because of a larger number of at-risk patients and surgical joint procedures in recent years.^{2,4–}

⁷ Estimates of the incidence in elderly patients and those with rheumatoid arthritis (RA) are much higher, ranging from 28 to 70 cases per 100,000 per year.^{2,4–8} Rates vary by ethnicity, and socioeconomically disadvantaged populations have a higher prevalence than the general population.^{2,3,7} Bacterial arthritis is an uncommon but not rare diagnosis in the emergency department (ED), accounting for 8% to 27% of patients presenting with one or more acutely painful joints and over 16,000 US ED visits in 2012 (0.01% of total).⁹ Approximately 80% of ED patients with septic arthritis are hospitalized.⁹ Information on the epidemiology of septic arthritis is limited by the retrospective nature and varying case definitions used for many of the studies and the lack of a reliable gold standard for the diagnosis.

The published 90-day mortality rates of bacterial arthritis in adults vary between 5% and 20% but may be as high as 30% to 50% in those with significant comorbidity, multiple joint involvement, or advanced age.^{2,3,7,10–16} A robust study from New Zealand included 543 cases of native joint septic arthritis and showed a higher incidence of large joint involvement than seen in previous studies.² Large joint septic arthritis was shown to occur more frequently in older patients with more comorbidities. Even with longer treatment courses, it still had higher treatment failure and mortality rates than small joint septic arthritis. Moreover, bone and joint infections that develop in a health care setting are associated with a significantly higher mortality rate, longer length of hospital stay, and greater financial cost than infections acquired in the community.^{16,17}

The morbidity of septic arthritis is considerable, with up to 50% of patients reporting permanent decreased joint function or mobility after infection, depending on the presence or absence of preexisting joint disease and identity of the infecting pathogen.^{13,17–20} Following acute management, arthroscopic surgery of the affected joint will be required in as many as 9% of individuals within 15 years, a rate six times higher than the general population.¹⁴ Despite improved antimicrobial agents and adjunct treatment measures, as well as advances in hospital care, the morbidity and mortality of septic

arthritis has not changed appreciably in the past two to three decades.

Route of Infection

Bacterial arthritis is usually hematogenously acquired during overt or occult bacteremia, including that caused by endocarditis and associated with intravenous drug use (IVDU).^{6,18,20,21} A study from Spain of cases from 1985 to 2011 of bacteremic osteoarticular infection noted an increase in cases, with an increased proportion of health care–related cases. Patients increasingly were found to be older, comorbidities were more common, and more cases were device related.²² Normal, diseased, and prosthetic joints are all susceptible to hematogenous infection, although abnormal joint architecture greatly increases the risk. Thus, patients with osteoarthritis, gout, and pseudogout are all at an increased risk of infectious arthritis.^{5,22} The extremely vascular synovial membrane of the joint lacks a limiting basement membrane and is particularly susceptible to the hematogenous deposition of bacteria.^{2,18,22,23} Most nosocomial cases of acute bacterial arthritis presumably occur via this route.^{15,24}

Other routes of infection include direct inoculation of bacteria into the joint through surgery, trauma, animal and human bites, percutaneous puncture (such as from a nail, needle, or thorn), or contiguous spread from adjacent infected soft tissue or bone.^{2,9} A common example of the latter is septic arthritis of the small joints of the foot, complicating an adjoining diabetic foot ulcer or infection.²⁵ On occasion, hip joint infection may occur from contiguous spread from an intraabdominal source via the retroperitoneal space and iliac acetabulum.^{26,27}

Septic arthritis is an unusual complication of arthroscopic surgical procedures of the knee, occurring in only 0.04% to 0.7% of arthroscopies and 0.14% to 2.3% of arthroscopic reconstructive procedures.^{28–30} The incidence of infection after arthroscopic surgery of the shoulder may be somewhat higher, with one study reporting rates of 8.5 per 1000 rotator cuff repairs.³¹ Although the risk of infection per procedure remains relatively unchanged, when compared to the previous decade a study from Iceland shows a rise in the incidence of native joint infections following arthroscopic

procedures from 9 cases to 25 cases per 100,000 persons per year due to more widespread use of arthroscopy.³² Similarly, it is extremely rare for infection to follow a single joint aspiration or corticosteroid injection, occurring in less than 0.04% of those procedures.^{5,33} Iatrogenic infections that follow joint corticosteroid injection may be caused by unusual pathogens and can occur in outbreaks or clusters related to contaminated medication, usually methylprednisolone.^{34,35} A recent practice trend of joint injection with hyaluronic acid compounds for management of osteoarthritis is also not without risk. A breakdown of infection prevention practices in such procedures was responsible for an outbreak of 41 cases of septic arthritis in New Jersey in 2017.^{36,37}

Predisposing Host Factors

Predisposing host factors for septic arthritis are listed in **Table 105.1**. The most important risk factors are preexisting abnormal joint architecture—particularly that caused by RA, and IVDU. A prospective cohort study of patients presenting with acute inflammatory arthritis found that recent joint surgery, age older than 80 years, diabetes mellitus, and RA were associated with a significant and increased incidence of septic arthritis (likelihood ratios: 6.9, 3.5, 2.7, and 2.5, respectively).¹ In the midst of the ongoing opioid epidemic, a study from Massachusetts of 1465 cases showed that IVDU is now the leading risk factor for septic arthritis.²¹ Prevalence was also increased for patients on hemodialysis in one study, although some of this risk was likely due to preexisting joint disease.³⁸ In one study ongoing tobacco smoking was found to be a significant risk factor for both small and large joint infection.²

The immunosuppressive disease-modifying antirheumatic drugs (DMARDs) penicillamine, sulfasalazine, and prednisone have been identified to further increase the risk in patients with RA.³⁹ Although methotrexate therapy was found to be associated with septic arthritis in one investigation,⁹ it was not so identified in a large retrospective controlled study.³⁹ Similarly, patients treated with anti-tumor necrosis factor (TNF) agents have increased risk for several infections, including those of the joint.^{39–44} A large prospective observational study found that anti-TNF therapy

doubled the risk (adjusted hazard ratio, 2.3) of septic arthritis in patients with RA. The risk was greatest in the first year of therapy and was not significantly different between the specific anti-TNF agents.⁴⁵ Moreover, opportunistic and intracellular pathogens, particularly mycobacteria and dimorphic fungi, are more likely to cause joint infections in these patients.^{44,46,47}Page 1335

TABLE 105.1
Predisposing Factors in Bacterial Arthritis

Major Factors
1. • Rheumatoid arthritis 2. • Crystal induced arthritis (gout and pseudogout) 3. • Advanced age 4. • Diabetes mellitus 5. • Chronic renal failure 6. • Ongoing smoking 7. • Previous joint surgery 8. • Penetrating joint injury 9. • Intravenous drug use 10. • Endocarditis 11. • Immunosuppression 1. • Organ and bone marrow transplant 2. • Immunosuppressive therapy, including systemic cor anti-TNF agents
Minor Factors

1. • Other joint disease
 1. • Osteoarthritis
 2. • Charcot arthropathy
2. • Chronic systemic disease
 1. • Collagen vascular disease
 2. • Malignancy
 3. • Chronic liver disease
 4. • Sickle cell disease
 5. • Alcoholism
3. • Hypogammaglobulinemia
4. • Intraarticular injection (e.g., glucocorticoids)
5. • Skin disease with or without infection
6. • Low socioeconomic status
7. • Aboriginal ethnicity

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Data from references 1–11, 20, 24, 25, 33, 34, 44, 49, 101, 117.

One study in Africa suggested that patients with human immunodeficiency virus (HIV) may have a small increased risk for septic arthritis,⁴⁸ whereas other studies have found no such association.^{49,50} The perceived increased risk in patients with HIV is likely related to concomitant IVDU.^{21,51,52}

An important source of hematogenously derived joint sepsis is skin disease or lesions, with or without concurrent infection.^{1,2,9} Of significance, up to 22% of patients with septic arthritis will not have an identifiable predisposing risk factor, and the negative predictive value of any one factor for joint infection is low. Thus the diagnosis must be considered in all persons presenting with inflammatory arthritis, even when risk factors are absent.^{11,53,54}

Nongonococcal Arthritis

Pathophysiology

The pathophysiology of acute nongonococcal septic arthritis is complex and dependent on adherence of organisms to, and colonization of, the synovial membrane, bacterial proliferation in synovial fluid, and a resultant synovial infection with generation of a host inflammatory response.⁵⁵ Much of our understanding of these

pathophysiologic mechanisms comes from studies of animal models of joint infection, particularly with *Staphylococcus aureus*.⁵⁶ After hematogenous or direct entry to the joint, bacterial adherence is facilitated by the low shear conditions of synovial fluid and, in some cases, by joint disease or injury (traumatic or surgical). The latter results in an increased amount or exposure of host-derived extracellular matrix proteins, such as fibronectin, collagen, laminin, elastin, and hyaluronic acid, which promote bacterial attachment.^{23,57} Some bacteria display a tropism for synovial infection, at least in part because of adherence characteristics of the organism. For example, *S. aureus* has surface receptors, such as fibronectin-binding protein and “microbial surface components recognizing adhesive matrix molecules” (MSCRAMMs), that recognize selected host proteins and mediate adherence to the joint extracellular matrix (see **Chapter 199**).^{58,59} Expression of these surface receptors, as well as several other virulence factors including lipoproteins, are influenced by genes, including those regulating quorum sensing, which promote *S. aureus* invasion and infection when host defenses are weakened.^{60–62} In the case of *Kingella kingae* and *Streptococcus agalactiae*, joint tropism and synovial adherence are influenced by bacterial pili and fibrinogen-binding adhesin proteins, respectively.^{57,63}

TABLE 105.2

Bacteria Isolated in 2302 Compiled Cases of Bacterial Septic Arthritis in Adults

ORGANISM	NO. OF ISOLATES (% OF
Gram positive	
<i>Staphylococcus aureus</i>	1066 (46)
<i>Staphylococci, coagulase n</i>	84 (4)
Streptococci	512 (22)
<i>Streptococcus pyogenes</i>	183 (8)
<i>Streptococcus</i>	156 (7)
<i>Streptococcus agalactiae</i>	69 (3)

Other streptococci	104 (5)
Gram negative	
<i>Escherichia coli</i>	91 (4)
<i>Haemophilus influenzae</i>	104 (5)
<i>Neisseria gonorrhoeae</i>	77 (3)
<i>Neisseria meningitidis</i>	28 (1)
<i>Pseudomonas aeruginosa</i>	36 (2)
<i>Salmonella</i> spp.	25 (1)
Other gram-negative rods	110 (5)
Miscellaneous (including	136 (6)
Polymicrobial	33 (1)

Modified from Ross JJ, Saltzman CL, Carling P, et al. Pneumococcal septic arthritis: review of 190 cases. *Clin Infect Dis*. 2003;36:319–327.

Although certain bacterial products or toxins may directly increase tissue damage in the infected joint,^{23,62,64–67} it is the host inflammatory response to infection that is responsible for much of the joint injury.^{68,69} In response to replicating bacteria and a number of bacterial products an inflammatory cell response is rapidly noted within the joint synovial membrane, which then responds with a proliferative lining-cell hyperplasia.⁷⁰ Selected host gene expression of proinflammatory or protective proteins has the potential to regulate this inflammatory cell response.⁷¹ An influx of acute and chronic inflammatory cells results in the characteristic purulent inflammation of the joint and its synovial fluid. Leukocyte-derived proteases and inflammatory cytokines, including interleukin-1 (IL-1), IL-4, IL-6, IL-10, and TNF- α , either directly or indirectly result in cartilage degradation, inhibition of cartilage synthesis, and subchondral bone loss.^{23,24,69,71,72} Intraarticular cartilage destruction may be seen in as little as 3 days.⁷³ In addition, the inflammatory joint infusion increases intraarticular pressure, hampering capillary blood flow to the joint, resulting in cartilage and synovial ischemia

and necrosis.¹⁸ Over time cartilage destruction leads to joint space narrowing and further erosive damage to the cartilage and underlying bone. If untreated, infection can spread from the joint to surrounding soft tissue, disrupting ligaments, tendons, and other periarticular structures and occasionally forming sinus tracts.^{74,75}

Microbiology

Septic arthritis is caused by a multitude of pathogenic bacteria (**Table 105.2**). Except for slight decreases in the proportion of cases due to gonococci, group A streptococci, and *Streptococcus pneumoniae* plus IVDU-related increases in *Staphylococcus aureus* and *Serratia* spp., the distribution of bacterial organisms causing septic arthritis has changed little over time.^{2–6,11,21,25,76–81}

Gram-Positive Bacteria

By far the most common etiology of septic arthritis in adults is *S. aureus*, which is responsible for 37% to 80% of cases, depending on geographic location, incidence of comorbid rheumatic disease, proportion of infections involving native joints, and rates of IVDU.^{21,82} For patients with RA the proportion of septic arthritis caused by *S. aureus* has been reported to be higher (~75%).^{74,83} Polyarticular infection with *S. aureus* in these patients is not uncommon.²¹ Page 1336

In many geographic locations the incidence of methicillin-resistant *S. aureus* (MRSA) septic arthritis has increased compared with that in the 1990s, particularly in the elderly, persons who inject drugs (PWID) or who have had recent orthopedic surgery, and those colonized or previously infected by MRSA.^{15,16,21,84–88} However, some studies, particularly those from Western Europe, have not noted this increase, and others have reported little change in the proportion of joint infections due to MRSA since 2000.^{25,80,89} Coincident with the increased prevalence of invasive infections caused by community-acquired MRSA (CA-MRSA) since 2003, there have been several reports of native joint infection caused by this pathogen, especially in children.^{16,89–95} Recent data, however, suggest that MRSA septic arthritis in children may be declining.⁹⁶ Septic arthritis caused by MRSA is associated with increased morbidity, suppurative complications, and duration of fever, antibiotics, and hospitalization, compared with that caused by

methicillin-susceptible *S. aureus* (MSSA).^{90,91,97} Vancomycin intermediate, but not high-level, resistance in *S. aureus* has rarely been reported as a cause of septic arthritis in patients with frequent exposure to health care facilities.^{95,98–100} Coagulase-negative staphylococci, a common pathogen in prosthetic joint infections, are rarely a cause of native joint septic arthritis. If isolated, however, it is important to differentiate *Staphylococcus lugdunensis* from other coagulase-negative staphylococci because of its increased virulence and propensity to cause invasive infections of native tissues.¹⁰¹ *Streptococcus* spp. are the next most frequently isolated bacteria from adults with native joint septic arthritis.^{4–6,23,25,76–80} *Streptococcus pyogenes* and other β -hemolytic streptococci from Lancefield groups C, F, and G are important pathogens within this group. *S. agalactiae* (Group B streptococcus or GBS) is a cause of bacterial arthritis in neonates, as well as adults with diabetes mellitus, malignancy, and genitourinary structural abnormalities, and, like *S. aureus*, it is prone to occur with polyarticular infection.^{102–}

¹⁰⁵ Population surveillance from 1995 to 2012 in California showed an increased incidence of septic arthritis due to this organism, with diabetes mellitus the most common underlying condition.¹⁰⁶ In addition, a retrospective review from 2006 to 2018 in Thailand showed GBS to be the most common pathogen in their patient population, with individuals less likely to have underlying comorbidities despite having more joint involvement and higher rates of concomitant tenosynovitis.¹⁰⁷ *S. pneumoniae*, traditionally thought of as a rare cause of hematogenous septic arthritis in the antibiotic era, accounted for 6% of cases in a comprehensive systematic review.⁷⁸ Polyarticular infection was frequent (36%), and many patients did not have prior or concurrent respiratory tract infection. Some studies have shown lower numbers of joint infections due to this organism, particularly in children, since the adoption of pneumococcal conjugate vaccine.⁸¹ However, despite the decreasing incidence of invasive pneumococcal disease in a 13 year cohort study from the United Kingdom, 13-valent pneumococcal vaccination shifted the percentage of pneumococcal joint isolates from serotypes included in the vaccine to nonvaccine serotypes.¹⁰⁸ Joint infections due to nonvaccine serotypes were associated with more severe disease.

Other streptococci are a less frequent cause of septic arthritis. *Streptococcus suis*, an emerging zoonotic pathogen in China, Southeast Asia, and, to a lesser extent, Europe, has been reported to cause septic arthritis in persons exposed to pigs or improperly cooked pork.^{109,110} Another zoonotic pathogen, *Streptococcus iniae*, is associated with aquaculture fish and has occasionally been described as a cause of bacterial arthritis in fish handlers.¹¹¹ Native joint septic arthritis caused by viridans group streptococci is rare, in contrast to prosthetic joint infections, where it is a common pathogen.¹¹² Osteoarticular infections with the *Streptococcus bovis* group, most frequently *Streptococcus gallolyticus* subsp. *gallolyticus*, are often a manifestation of infectious endocarditis, and evaluation for endocarditis and colonic malignancy should be undertaken.¹¹³

Gram-Negative Bacteria

Gram-negative bacilli are cultured from approximately 5% to 20% of patients with bacterial arthritis, particularly from neonates, the elderly, PWID, and immunocompromised hosts.^{5,6,23,80,114,115} The coliform bacteria are most commonly isolated from the elderly, immunocompromised, and comorbidly ill^{2,25,77,79} and are increasingly found to be multidrug resistant.^{116,117} *Pseudomonas aeruginosa* is an important pathogen in PWID and has also been described as a cause of iatrogenic septic arthritis after surgical procedures and intraarticular injections.^{118,119} It has a particular affinity for fibrocartilaginous articular structures, such as the pubic symphysis and the sternoclavicular, sternochondral, and sacroiliac joints.^{120–123} *Haemophilus influenzae*, once an important pathogen in young children, is a rare cause of septic arthritis in populations where *H. influenzae* type b vaccine is widely used.¹²⁴ Although uncommon, nontypeable *H. influenzae* joint infections still occur. In a retrospective review of 121 cases of pediatric septic arthritis it was the third most isolated organism.¹²⁵ Most of these infections were in children of Native American descent.

For children younger than 4 years, *Kingella kingae*, a resident of the normal oral flora, has replaced *H. influenzae* as the principal gram-negative cause of hematogenous bacterial arthritis.^{126–130} Joint infection caused by *K. kingae* is often associated with stomatitis or upper respiratory tract infection and typically responds well to

therapy with minimal articular or functional sequelae. Child-to-child transmission of this communicable bacterium occurs, and outbreaks of *K. kingae* osteoarticular infections in childcare centers have been reported.^{130–132}

Other Bacteria

Other nongonococcal bacteria identified in infected joints include corynebacteria, *Salmonella* spp., *Neisseria meningitidis*, *Listeria monocytogenes*, *Burkholderia cepacia*, *Burkholderia pseudomallei*, *Legionella* spp., and *Brucella* spp. Articular infection caused by anaerobic bacteria is rarely reported.¹³³ An important exception, however, is *Cutibacterium acnes* (formerly *Propionibacterium acnes*), which is often isolated from infected shoulder joints after arthroscopic or open surgery.^{31,134,135}

Unusual bacterial pathogens causing contiguous septic arthritis after a dog or cat bite include *Pasteurella multocida* and *Capnocytophaga* spp., and, in the case of a human bite, *Eikenella corrodens* and *Fusobacterium nucleatum*.^{136–}

¹³⁹ *Streptobacillus moniliformis*, a causative agent of rat-bite fever, is occasionally isolated from the blood or synovial fluid in patients with polyarticular arthritis after a rat bite.^{140,141} *Pantoea*

agglomerans is the most common bacteria cultured from infected joints after direct plant thorn injury.¹⁴² **Table 105.3** lists selected clinical and epidemiologic associations with septic arthritis and their likely causative bacteria.

Cultures of synovial fluid or blood from patients diagnosed with septic arthritis yield polymicrobial flora in up to 10% of cases.

Patients are sicker at presentation and have higher rates of intensive care unit–level care and longer hospital stays.^{25,77,143} Causes of bacterial arthritis in which a pathogen is not isolated from blood or joint fluid using conventional culture techniques are *Mycoplasma hominis*, *Ureaplasma urealyticum*, *B. burgdorferi*, and *Tropheryma whippelii* (Whipple disease).

Clinical Manifestations

Joints Involved

Nongonococcal septic arthritis is monoarticular in 80% to 90% of cases, with the knee being the predominant site of infection in up to 50% of patients.^{2,5–7,20,21,25,32} Other native joints that are frequently

involved in adults include the hip, shoulder, wrist, and ankle. In children hip infections predominate.¹⁴⁴ Infections of the peripheral joints of the hands are unusual, except in the setting of trauma, particularly animal or human bites.^{2,138} Septic arthritis of the small joints of the foot is most often secondary to contiguous spread from skin and soft tissue ulcerations or adjacent osteomyelitis and is mostly seen in diabetic patients.^{2,145} Nongonococcal polyarticular bacterial arthritis is observed in 10% to 20% of patients with septic arthritis, especially in those with RA, immunosuppression, or prolonged or intense bacteremia that is often associated with endovascular infections and IVDU.^{2,5,6,21} These infections are usually caused by *S. aureus*. Acromioclavicular, sternoclavicular, costochondral, or facet joint infections are uncommon except in PWID and occasionally as a complication of overt or occult bacteremia.^{2,125,146–150} In these cases *P. aeruginosa* is a frequent pathogen. Risk factors for septic arthritis of the pubic symphysis include female urinary incontinence surgery, participation in athletics, pelvic malignancy, and IVDU.^{23,121,122}Page 1337

TABLE 105.3

Clinical and Epidemiologic Features Associated With Selected Bacterial Causes of Septic Arthritis

CLINICAL OR EPIDEMIOLOGIC FEATURE	ETIOLOGIC AGENT
Rheumatoid arthritis	<i>Staphylococcus aureus</i>
Intravenous drug use	<i>S. aureus</i> , <i>Pseudomonas</i>
Diabetes, malignancy	<i>S. aureus</i> , <i>Streptococcus</i>
Immunocompromised hosts	<i>S. aureus</i> , streptococci, enteric negative bacilli, <i>Listeria monocytogenes</i>
Neonates, children younger than 4years	<i>S. agalactiae</i> , gram-negative bacilli, <i>Kingella</i> <i>kingae</i> , nontypeable <i>Haemophilus</i> <i>influenzae</i>
Young adults, menstruating females,	<i>Neisseria gonorrhoea</i>

Fibrocartilaginous joints (e.g., pubic symphysis)	<i>S. aureus</i> , <i>P. aeruginosa</i>
Postsurgical shoulder	<i>Cutibacterium acnes</i>
Sacroiliac joint	<i>S. aureus</i> , <i>Salmonella</i> spp., <i>Listeria</i>
Cat or dog bite	<i>Pasteurella multocida</i> , <i>Capnocytophaga</i> spp., anaerobes
Human bite	<i>Eikenella corrodens</i> , anaerobes, oral flora (e.g., viridans streptococci)
Rat bite	<i>Streptobacillus</i>
Postpartum women	<i>Mycoplasma hominis</i>
Ingestion of unpasteurized dairy products, residents or	<i>Brucella</i> spp.
Residents or travelers to Southeast Asia	<i>Burkholderia pseudomallei</i> (melioidosis), <i>B. mallei</i> , <i>B. thailandensis</i>
After plant thorn injury	<i>Pantoea agglomerans</i> , <i>Nocardia</i>

Sacroiliac joint infection is an uncommon metastatic complication of occult or symptomatic bacteremia, usually caused by *S.*

aureus.^{120,146,151–156} PWID and patients with indwelling catheters are at risk. Patients typically present with acute onset of sacral or pelvic pain and leukocytosis. Blood cultures are often positive. Patients should be evaluated for the presence of concomitant infective endocarditis. Uncommon causes include *Salmonella* spp., including *Salmonella enterica* serovar Typhi, and brucellosis.^{152,154,155,157} Brucellosis causes a chronic unilateral infection of the sacroiliac joint that must be distinguished from arthritis associated with inflammatory bowel disease (IBD) and ankylosing spondylitis (see **Chapter 231**). Injection of contaminated methylprednisolone into the sacroiliac joint led to an indolent sacroiliac joint infection caused by *Exserohilum* during a multistate outbreak of meningitis, epidural abscess, and joint infection caused by this pathogen.¹⁵⁶

Clinical Presentation

Most patients with acute bacterial arthritis present with the cardinal symptoms of arthritis—pain and loss of function of one or more joints over a 1- to 2-week period. In addition to intense pain and decreased range of motion, other symptoms of nongonococcal bacterial arthritis include swelling, redness, and increased warmth of the infected joint.^{3,6,11,18,24} Joint pain is often the only focal symptom of deep or axial joint infection. In the case of sacroiliac or pubic symphysis septic arthritis, pain is intensely exacerbated by ambulation and often radiates to the back or groin, respectively, and to the hip and proximal leg in either case.^{120,122,158,159} Although fever and malaise are commonly associated with bacterial arthritis in adults, high fever with rigors and shaking chills are typically not apparent and may be totally absent in 30% to 50% of patients.^{6,18,20,24} A systematic review showed that fever was present as a physical examination finding in approximately 50% of patients, and it was found to have a sensitivity for articular infection of only 57%.⁵³

Physical examination findings of infected peripheral joints include focal joint tenderness, inflammation, and effusion (**Fig. 105.1**). Active and passive range of motion of the joint is usually limited and results in considerable discomfort. Examination findings in patients with bacterial arthritis of the nonperipheral joints may be limited to focal tenderness over the afflicted area. Children with septic arthritis of the hip characteristically hold the hip in a flexed and externally rotated position and resist any range of motion. Weight bearing is painful. A source of infection distant from the afflicted joint, including that of the skin, may be discovered after careful clinical evaluation in up to 50% of patients.²⁰



FIG. 105.1 (A) Bacterial arthritis of the knee showing swelling and effusion. (B) Septic third proximal interphalangeal joint demonstrating striking redness, swelling, and effusion.

[Follow for extended description](#)

Patients who are immunocompromised; have comorbid illnesses, including RA; or are at the extremes of age often present without a fever and with significantly more subtle symptoms and signs of joint infection, frequently resulting in delayed diagnosis.^{3,4,114,158} For patients with RA or other chronic joint disease, particularly those receiving systemic or intraarticular steroids, manifestations of a complicating joint infection may be limited to a modest increase in the severity of chronic articular inflammation and loss of joint function. In these cases, it can be very difficult to differentiate joint infection from an exacerbation of their underlying disease.^{10,45,159}

Laboratory Findings

Laboratory evaluation frequently shows an elevated peripheral white blood cell (WBC) count, erythrocyte sedimentation rate (ESR), and C-reactive protein (CRP); however, these tests are nonspecific and, if elevated, do not appreciably raise the pretest probability of septic arthritis.^{24,53,160,161} Similarly, a nonelevated ESR or CRP may be found in native joint infections due to low virulence organisms.^{162,163} One retrospective analysis showed a CRP exceeding 10.5 mg/dL was predictive of septic arthritis.¹⁶⁰ Other studies suggest that CRP levels >20 mg/dL are a reliable independent marker to differentiate septic arthritis from crystal arthropathy and osteo/inflammatory arthritis and that it remains valid even in immunocompromised hosts.^{163,164}

Serum procalcitonin (PCT), an acute-phase reactant more sensitive and specific than ESR and CRP, may potentially be useful for the diagnosis of acute bacterial arthritis.^{165,166} Serum PCT was evaluated in a meta-analysis for its usefulness as a diagnostic marker for septic arthritis.¹⁶⁷ For those patients with septic arthritis, using a cutoff level of 0.5 ng/mL, the diagnostic sensitivity and specificity were 54% and 95%, and the positive and negative likelihood ratios were 11.0 and 0.5, respectively. Furthermore, serum PCT was noted to be a more useful indicator of septic arthritis than CRP.¹⁶⁷ Adjusting the cutoff level for positivity gives modestly different performance characteristics, and there is no clear consensus as to the optimum value.¹⁶⁸ Thus an elevated serum PCT might help suggest bacterial septic arthritis in a patient with an acutely inflamed joint, but, more important, a normal value is not useful for ruling it out.^{165,167,169}

Arthrocentesis of an affected joint usually reveals purulent, low-viscosity synovial fluid with an elevated polymorphonuclear neutrophil count. Traditionally, a synovial fluid leukocyte count greater than 50,000 cells/mm³ has been used to suggest a diagnosis of septic arthritis^{2,4,18}; however, lower WBC counts are regularly encountered and do not exclude the diagnosis (**Table 105.4**).^{21,45} Synovial fluid WBC counts between 25,000 and 75,000 cells/mm³ constitute a “gray zone” for the diagnosis of bacterial arthritis, and careful clinical assessment and observation is necessary in such patients.¹⁷⁰ Of note, administration of antibiotics within 2 weeks prior to arthrocentesis alters synovial fluid WBC counts and neutrophil percentages, in which case a cutoff of 16,000 cells/mm³ may be optimal in evaluation for these patients.¹⁷¹ Other potential tests of synovial fluid laboratory tests have been evaluated. Although synovial fluid glucose may be depressed and protein and lactate dehydrogenase elevated, the sensitivity and specificity of these tests are low and limit their usefulness.⁵³ A single study evaluating the synovial lactate-glucose ratio for diagnosis showed higher performance for differentiating septic arthritis from non-septic arthritis than the synovial lactate or glucose assays individually, but further research is needed.¹⁷² The diagnostic accuracy of an elevated synovial fluid lactic acid, PCT, or an inflammatory cytokine, such as TNF-α, has not been adequately assessed.^{24,53,173–175}

TABLE 105.4
Diagnostic Parameters of Synovial Fluid White Blood Cell Count and Percent Polymorphonuclear Cells in Patients With Septic Arthritis

	SENSITIVITY (%)	SPECIFICITY (%)	LIKELIHOOD RATIO (95% CI)	
			Positive	Negative
>100,000 WBC/mm ³	29	99	28.0 (12.0–65.0)	0.71 (0.64–0.79)
>50,000 WBC/mm ³	62	92	7.7 (5.7–10.5)	0.42 (0.34–0.51)

>25,000 WBC/mm³	77	73	2.9 (2.5–	0.32 (0.23–
PMN cells	73	79	3.4 (2.8–	0.34 (0.25–

CI, Confidence interval; *PMN*, polymorphonuclear; *WBC*, white blood cell.

Modified from Margaretten ME, Kohlwes J, Moore D, et al. Does this adult patient have septic arthritis? *JAMA*. 2007;297:1478–1488.

Microbiologic Diagnosis

Conventional culture

Synovial fluid culture in nongonococcal infection of patients who have not previously received antibiotics will yield bacterial growth up to 80% to 90% of the time; however, Gram staining is diagnostic in only 50% of these cases.^{12,79,161} False-positive Gram stains of synovial fluid can occur because of artifacts from stain, mucin, and cellular debris. There is some evidence that direct inoculation of synovial fluid into pediatric or adult blood culture media bottles or isolator culture tubes improves the recovery of pathogens,^{176,177} although one investigation comparing the BACTEC (Becton Dickinson, Franklin Lakes, NJ) blood culture bottle to lysis centrifugation and conventional agar plate culture techniques found similar yields from all three methods.¹⁷⁸ However, using a blood culture system for synovial fluid culture may enhance the yield of fastidious organisms, such as *K.*

kingae and *Brucella* spp.^{179,180} Synovial fluid cultures from potentially infected shoulders should be incubated for at least 10 days, to enhance the isolation of *C. acnes*, a slow-growing bacterium that is an important cause of infection of this joint.¹⁸¹

The proportion of patients with positive blood cultures in nongonococcal bacterial arthritis varies from 25% to 70%, depending on the study.^{13,20} Of note, they are the sole source of the infecting organism's identity in approximately 10% of cases. Both synovial fluid and blood culture sensitivity decline in patients after antimicrobial therapy has been initiated.

False-positive synovial fluid cultures due to bacterial contamination occasionally occur and are more likely in aspirates from the wrist and shoulder. Clinical features suggesting a false-positive culture include low synovial fluid WBC counts, isolation of coagulase-negative *Staphylococcus* or *Bacillus* spp., and time to positivity of more than 48 hours.¹⁸² *C. acnes* can be especially problematic when isolated from a native or prosthetic shoulder joint. Although this organism is a significant cause of infection of this joint, false-positive cultures also occur. Growth of the organism after 9 days of incubation is more likely to be a culture contaminant; however, careful clinical correlation is essential before discounting it as a false-positive culture.¹⁸³

Molecular tests

Identification of bacterial DNA in synovial fluid by multiplex or broad-range polymerase chain reaction (PCR) assay shows promise for improving the etiologic diagnosis of septic arthritis, especially for infections caused by fastidious or unusual pathogens.^{175,184–}

¹⁸⁹ Identification of *K. kingae*, *Streptococcus* spp., *Mycoplasma*, and anaerobic bacteria is particularly enhanced. One technique showed a sensitivity and specificity of 95% and 97%, respectively, by using a 16S ribosomal DNA (rDNA) gene target and real-time PCR.¹⁸⁵ The negative predictive value in a separate study was 98%.¹⁹⁰ Addition of PCR testing to culture in another study enhanced the sensitivity for diagnosis of nonstaphylococcal bone and joint infection by 10% as well as spondylodiscitis due to *Mycobacterium tuberculosis* by 12%.¹⁹¹ Although PCR performance is generally thought to be high, one study comparing three 16S rDNA targets reported more modest results, with sensitivities of 44% to 63% and negative predictive values of 84% to 89%.¹⁹² An additional study showed that 16S rDNA testing had lower diagnostic performance than Gram stain, blood cultures, and synovial cultures.¹⁹³ Furthermore a study using multiplex PCR testing reported a negative result by PCR for *S. aureus*, *S. pneumoniae*, and *S. dysgalactiae* (GBS) despite positive synovial fluid cultures for these organisms.¹⁹⁴

Importantly, a multicenter evaluation of a US Food and Drug Administration (FDA) recently cleared multiplex PCR for bone and joint infection showed positive percent agreement and negative percent agreement of 91.6% and 93.0%, respectively, compared to

bacterial culture.¹⁹⁵ There were notable improvements in diagnostic yield, identification of polymicrobial infections, and detection of microbial resistance genes. A retrospective review at a large health system of this multiplex PCR in synovial fluid further showed that implementation could lead to earlier optimized therapy in as many as two-thirds of patients.¹⁹⁶

In addition to the identification of fastidious organisms, other benefits of PCR include rapid results, high negative predictive values, and greater likelihood of yielding a pathogen from patients who have previously received antibiotics. Limitations of PCR include false-positive tests as a result of sample contamination, lack of antimicrobial susceptibility data, and, in the case of broad-range PCR, difficulty in confirming a specific species because of shared regions of bacterial 16S DNA.

Synovial fluid metagenomic next-generation sequencing allows for identification of a larger pool of genome-known organisms and is increasingly being used for study in adult and pediatric native joint septic arthritis and periprosthetic infections.^{197–199} One such study of operative specimens in pediatric osteoarticular infections showed metagenomic next-generation sequencing enabled identification of a causative pathogen in four cases that were negative on routine culture and PCR testing.¹⁹⁷ Overall positive and negative predictive agreement was 93%. Synovial fluid sequencing shows promise, but further study is needed to assess optimal clinical applications.

In summary, molecular testing as a tool for the diagnosis of septic arthritis is becoming increasingly available and useful for the etiologic diagnosis of septic arthritis.

Radiographic Features

Imaging studies of joints and periarticular structures affected by bacterial arthritis often yield additional information helpful for establishing the diagnosis or evaluating for complications of infection.²⁰⁰ Plain film radiography in early infection will typically show periarticular soft tissue swelling and fat-pad edema but normal osseous structures.²⁰¹ In advanced infection findings may include periarticular osteoporosis, joint space loss, periosteal reaction, marginal and central erosions, and destruction of subchondral bone. Plain films are useful for assessment of preexisting joint disease, metallic foreign bodies, and simultaneous

osteomyelitis or as a baseline image in monitoring for sequelae of infection.

Scintigraphy, computed tomography (CT), and magnetic resonance imaging (MRI) are highly sensitive techniques for imaging early septic arthritis. Scintigraphy, including three- or four-phase bone scanning, Page 1339 using technetium 99m methylene diphosphonate is a sensitive but nonspecific indicator of articular infection.²⁰¹ CT is useful for the detection of erosive bone changes, subchondral cysts, joint effusions, and periarticular soft tissue extension of infection, particularly in deep articulations, such as the hip, sacroiliac, and sternoclavicular joints and in the pubic symphysis (**Fig. 105.2**).^{122,157,201–203}

Similar to bone scanning and CT, MRI is an extremely sensitive procedure for the evaluation of bacterial arthritis but is more specific than these other modalities.^{204,205} MRI is particularly valuable for the detection of joint inflammation, effusion, and articular cartilage destruction, as well as for periarticular cellulitis, fistula, abscess, or osteomyelitis of contiguous bone. In some cases findings on MRI may discriminate between septic arthritis and underlying chronic RA.²⁰⁶ However, MRI may pose difficulties distinguishing septic arthritis from sterile inflammatory arthropathies, cannot be used with certain metal joint implants, and has lower resolution for the bony cortex than CT.^{23,204} Dynamic contrast-enhanced MRI, a technique that has shown usefulness in differentiating septic arthritis and transient tenosynovitis involving the hip joint, may also have use for evaluation of other joints.²⁰⁷ In most US medical centers, MRI and CT have replaced scintigraphy as the preferred imaging modalities for evaluation of a potentially infected joint. Fluorodeoxyglucose-positron emission tomography (FDG-PET CT) offers no advantage over conventional CT/MRI imaging for evaluation of septic arthritis or differentiation from other causes of inflammatory arthritis.²⁰⁸

Particularly in children, sonography of the hip can provide rapid evidence of effusion and can guide prompt joint aspiration without waiting for advanced imaging.^{200,209,210} A study evaluating functional outcomes associated with ultrasound abnormalities in septic arthritis in adults showed that persistent or higher Doppler grade on sonography at 2 weeks predicts motion range limitation at 3

months. No evidence was seen that persistent synovitis or joint effusion on ultrasound at 3 months correlated with treatment failure.²¹¹

Gonococcal Arthritis

Epidemiology

During the 1970s and 1980s *Neisseria gonorrhoeae* was the predominant cause of bacterial arthritis in sexually active adults and adolescents, with the majority of cases occurring in persons younger than 30 years.^{212,213} The prevalence of gonococcal arthritis markedly declined as the rate of mucosal gonorrhea decreased fivefold between 1975 and 2011 to one of the lowest rates since statistics have been recorded.²¹⁴ Since 2011, however, rates of mucosal gonorrhea have rebounded, and consequently the prevalence of gonococcal arthritis has likely followed this increase.²¹⁵

Gonococcal arthritis is one of two clinical manifestations of disseminated gonococcal infection (DGI), the other being a syndrome of tenosynovitis, dermatitis, and polyarthralgia without purulent joint infection.^{212,216} There is overlap between the two conditions, and in some patients disseminated infection may progress from a bacteremic tenosynovitis-dermatitis syndrome to localized joint infection. Septic monoarticular or oligoarticular arthritis occurs in 42% to 85% of patients with disseminated gonococcal infection.²¹⁷

DGI is usually diagnosed in persons younger than 40 years, is approximately four times more common in women than in men, and complicates 0.5% to 3% of persons with mucosal gonococcal infection.^{212,218} More recently the proportion of cases occurring in men is likely increasing due to the increasing percentage of mucosal infection in men with HIV and men who have sex with men (MSM).^{215,219} Epidemiologic characteristics associated with DGI consist of lower socioeconomic status, non-White ethnicity, MSM, multiple sexual partners, and illicit drug use.²¹⁶ The incidence is markedly less in Europe than in North America, and it is considerably higher in the developing world, including in travelers to non-Western countries.²¹² A retrospective study from France showed that 41% of cases of gonococcal arthritis were associated with travel to Madagascar and Southeast Asia.²²⁰

Pathogenesis

Gonococcal arthritis and DGI occur as a result of occult bacteremia secondary to mucosal infection of the urethra, uterine cervix, rectum, or oropharynx. Asymptomatic mucosal infection is much more likely to result in DGI than symptomatic infection and may have been sexually contracted days to months before dissemination. This, at least in part, accounts for the increased risk of disease in women for whom endocervical gonorrhea may persist without symptoms for extended periods. For these women the risk of DGI increases substantially during menstruation, pregnancy, and the direct postpartum period. One-third to one-half of all women who present with DGI will do so during these times.^{217,221,222} Other host factors that increase the risk of DGI are complement deficiencies, particularly the terminal components C5 to C8,^{223–226} and systemic lupus erythematosus.²²⁷ Patients receiving eculizumab, a monoclonal antibody that inhibits terminal complement activation and thus increases risk for *Neisseria meningitidis* infection, also appear to be at higher risk for DGI.²²⁸ Immune mechanisms, including circulating immune complexes, also likely play a part in the pathogenesis of DGI and gonococcal arthritis and may account, in part, for the low yield of cultures for *N. gonorrhoeae* in synovial fluid and skin pustules.²¹³



FIG. 105.2 (A) Septic arthritis of the symphysis pubis. Computed tomography of the pelvis reveals joint space widening (*arrow*) with subchondral bone resorption and disruption of the articular cortical margins. (B) Septic sacroiliac joint. Coronal T2-weighted magnetic resonance image demonstrates fluid within the right sacroiliac joint, spreading superiorly and inferiorly to form soft tissue abscesses. *From Chew FS, Maldjian C, Leffler SG. Musculoskeletal Imaging: A Teaching File. Philadelphia: Lippincott Williams & Wilkins; 1999:267.*

[Follow for extended description](#)

N. gonorrhoeae possesses several virulence factors, many of them cell surface proteins, that influence pathogenesis and the ability of

the organism to widely disseminate from infected mucosa. Microbial attachment to mucosal and synovial epithelium is facilitated by long pili that also play a role in inhibiting host leukocyte phagocytosis.²²⁹ Gonococcal strains with the ability to disseminate are serum resistant and almost always express protein 1A, a principal outer membrane protein.^{218,230–232} Other characteristics of strains with an increased propensity to disseminate include lack of outer membrane protein II, also called “opacity protein,” and nutritional requirements for arginine, hypoxanthine, and uracil.^{23,212,233} *N. gonorrhoeae* strains that cause DGI are resistant to the bactericidal effects of normal human serum, probably through changes in cell membrane lipooligosaccharide.²³⁴

Antimicrobial Resistance

N. gonorrhoeae displays acquired resistance to several antimicrobials, including β -lactams, tetracyclines, fluoroquinolones, and, rarely, spectinomycin.^{235,236} The rapid emergence of fluoroquinolone resistance in *N. gonorrhoeae* throughout the world since 2001 has resulted in fluoroquinolones being contraindicated for empirical therapy of gonococcal infection.^{235–238} In addition, minimal inhibitory concentrations (MICs) of the oral cephalosporin cefixime against *N. gonorrhoeae* from North America increased markedly between 2009 and 2011.²¹⁵ This, together with treatment failures using cefixime for gonococcal mucosal infections, led the Centers for Disease Control and Prevention (CDC) to no longer recommend this treatment for gonorrhea in 2010.^{239,240} Amplifying the problem, MICs of azithromycin, the remaining oral drug with activity against *N. gonorrhoeae*, have also sharply increased. Between 2012 and 2020 the proportion of isolates with elevated MICs to azithromycin steadily increased from 0.3% to 5.8%, with the proportion higher among MSM populations.²¹⁵ As a consequence, ceftriaxone, the only antimicrobial with reliable activity against *N. gonorrhoeae*, is the sole remaining cornerstone for treatment of gonorrhea. Treatment guidelines updated in 2020 support the use of high dose ceftriaxone (500 mg) given the increasing prevalence of pharyngeal gonorrhea and the variability of pharmacokinetic and pharmacodynamic properties of the drug in the population.^{237,241}

Fortunately, in 2020 only 0.1% of isolates showed decreased susceptibility to ceftriaxone, and no isolates with elevated azithromycin MICs had elevated ceftriaxone MICs. An investigation from China of 1686 patients with uncomplicated gonorrhea showed no treatment failures with ceftriaxone despite 9.8% of isolates showing reduced susceptibility on testing. However, dosing regimens of 1 g were used in most patients, which may confound results.²⁴² Regrettably, precise epidemiologic data are not available on the antimicrobial susceptibility of strains of *N.*

gonorrhoeae isolated from patients with DGI or gonococcal arthritis compared with strains from mucosal infections. One report showed no resistance to ceftriaxone in a series of 58 cases of gonococcal arthritis from France from 2008 to 2020.²²⁰

This steady emergence of resistance of *N. gonorrhoeae* to numerous antimicrobials, particularly in the Pacific Rim and among MSM, has led to considerable concern that gonorrhea could soon attain an untreatable status.^{240,243} In January 2023 the Massachusetts Department of Public Health isolated a novel strain of gonorrhea from a patient with urethritis that showed reduced susceptibility or resistance to all antibiotics recommended for treatment. Although still successfully treated with ceftriaxone, this represents the first case of multidrug-nonsusceptible gonorrhea in the United States and further adds to the growing concern of drug resistance in this organism.²⁴⁴ Zoliflodacin has been evaluated as a new agent for the treatment of multidrug resistant *N. gonorrhoeae*.²⁴⁵

Clinical Manifestations

Patients with DGI typically present with the classic triad of dermatitis, tenosynovitis, and migratory polyarthralgia or polyarthritis. Joint symptoms may be quite severe and are often asymmetrical. Moderate fevers, chills, and general malaise are usually present. Two-thirds of patients develop tenosynovitis, usually occurring in the fingers, hands, and wrists, although the distal large and small joints of the lower limbs can also be involved (**Fig. 105.3**). The hip is rarely affected. Less than one-half of patients will have a true septic arthritis with a purulent joint effusion. The lesions of dermatitis are seen in two-thirds of DGI patients, are painless and nonpruritic, are few in number, and may not be noticed by the patient.^{216,246} Macules, papules, and pustules on

an erythematous base are most commonly seen and will often develop central necrosis (**Fig. 105.4**). The rash characteristically appears simultaneously with tenosynovitis and will be seen in various stages of development involving the limbs and hands (including the palms and soles), less often the torso, and rarely the face. Lesions will resolve after 4 to 5 days even if untreated, and new lesions may materialize after antimicrobial therapy is initiated.²¹⁶



FIG. 105.3 Photograph showing tenosynovitis of the dorsal hand accompanying monoarticular arthritis of the wrist. *Neisseria gonorrhoeae* was isolated from synovial fluid culture of the wrist.

Follow for extended description



FIG. 105.4 Clinical image demonstrating a pustular lesion overlying the fifth toe in a patient with disseminated gonococcal infection. *Courtesy Graham International Dermatopathology Learning Center.*

Follow for extended description

Septic gonococcal arthritis arising without tenosynovitis or skin lesions is less common than the so-called bacteremic form described previously and is clinically indistinguishable from bacterial arthritis caused by other organisms. The knees, wrists, and ankles are most commonly affected, and monoarthritis is more common than polyarthritis. Joints tend to be markedly inflamed, with large effusions. A syndrome of tenosynovitis with skin lesions may have occurred before the patient seeks medical care. Mucosal infection is usually present in patients with DGI and purulent gonococcal arthritis, although it is usually asymptomatic.

Laboratory Testing

As with nongonococcal septic arthritis, the laboratory findings of gonococcal arthritis and DGI include mild leukocytosis and elevated ESR in about 50% of patients.^{216,217} Synovial fluid from joint effusions afflicted with gonococcal arthritis will frequently, but not invariably,

have 50,000 to 100,000 WBCs/mm³, predominantly neutrophils, whereas joint aspirates from patients with DGI without frank suppurative arthritis will exhibit lower cell counts.²¹⁶ *N.*

gonorrhoeae is cultured from synovial fluid in approximately 50% of these cases and in 20% to 30% of patients with DGI.^{212,216,217} Only 25% of Gram-stained joint aspirates from patients with suppurative gonococcal arthritis reveal intracellular and extracellular gram-negative diplococci. Significantly higher proportions of patients with DGI are found to have *N.*

gonorrhoeae identified in synovial fluid by PCR assay than by conventional culture, with PCR assay sensitivity reported around 80%.²⁴⁷⁻²⁴⁹ Synovial fluid PCR for *N. gonorrhoeae* has a growing body of evidence supporting its use as a diagnostic aid.^{191,195} It is currently a component of a recently FDA-cleared multiplex PCR panel for bone and joint infections that shows improved diagnostic yield over conventional cultures.¹⁹⁵ Thus PCR testing may allow for earlier targeted therapy of gonococcal arthritis.¹⁹⁶

For patients with DGI and gonococcal arthritis, culture yield of *N. gonorrhoeae* is much higher from mucosal sites than from synovial fluid or blood. Cultures are positive from uterine cervical swabs in 80% to 90% of women and from urethral swabs in 50% to 70% of men.^{216,217,221} Culture yields from rectal and oropharyngeal swabs are lower but will sometimes give positive results. Positive blood cultures are exhibited by less than 30% of patients and are more common in those with dermatitis and tenosynovitis.^{246,250,251} Gram staining and bacterial culture of skin lesions rarely yield organisms. The use of nucleic acid amplification tests (NAATs) that are cleared by the FDA are now the recommended diagnostic testing modality for the detection of genital tract infection in women and men.^{237,252} The performance of NAATs with respect to overall sensitivity, specificity, and ease of specimen transport is better than that of other tests available for the diagnosis of gonococcal and chlamydial infections.^{253,254} Thus a mucosal NAAT should be performed for anyone with suspected gonorrhea, DGI, and gonococcal septic arthritis. NAATs should be obtained from a pharyngeal swab, a rectal swab, and either a vaginal swab, physician-collected endocervical swab in women and transgender men, or a first-catch urine specimen depending on individual site

exposure risk. Swabs for pharyngeal, rectal, and vaginal specimens can be collected by the patient or a clinician.^{239,254} A study from 2020 using a master protocol to analyze the performance of multiple NAATs for extragenital infection of both gonorrhea and chlamydia showed positive agreement of 94% to 96.5% and negative agreement of 98.8% to 99.6%, ultimately leading to FDA clearance of NAAT for extragenital testing.²⁵⁵ When treatment failure is suspected, cultures should be performed in addition to NAAT for specimens from all sites for antimicrobial susceptibility testing.^{237,252}

Septic Arthritis in Persons Who Inject Drugs

IVDU and addiction have been well described for more than 50 years, but since 2010 it has become a significant public health crisis, particularly opioid use disorder (OUD). In 2020, 9.5 million Americans were users of prescription opiates or IV heroin.²⁵⁶ Concomitant with this outbreak of OUD, nonviral IVDU infections have increased markedly in PWID, resulting in substantial morbidity, mortality, and increased hospitalizations in this population.^{257,258} (See **Chapter 316**.)

Although accurate numbers are not available, bone and joint infections in PWID are common. Estimates of the proportion of all such infections presenting in either identified or later-identified PWID increased from 3% to 12% in 2000, to 11% to 15% in 2014, and then to 20% in 2009-2018. Incidence is higher in younger age groups.^{21,257,259} IVDU is now the most common attributable risk factor for septic arthritis in adults.²¹ Up to half of all PWID-related joint infections may be in unrecognized users; thus there are opportunities to diagnose or treat underlying drug use in patients presenting with septic arthritis, particularly those who are of younger age.²⁵⁷

Morbidity, mortality, and complications of hospitalization are higher for septic arthritis in PWID than in those who do not inject drugs.²¹ One report found an adjusted odds ratio for death during hospitalization of 2.86 and repeat arthroscopic or open débridement of 1.24 and 1.68, respectively. In addition, they were more likely to leave against medical advice and have an additional 5 days of hospital stay.²⁵⁹

The bacteria responsible for joint infections in PWID are acquired either from commensal flora from the user's skin or saliva, or from organisms contaminating illicit drugs, drug diluents/adulterants, or paraphernalia.^{260,261} Occult or overt bacteremia is the most common route for joint seeding. Although commensal skin flora is the most usual origin of pathogens, contaminated water is also a frequent source. Depending on the study, *S. aureus* accounts for 50% to 90% of joint infections with methicillin-resistant isolates accounting for 18% to 40% of these.^{21,260–263} *S. aureus* and MRSA are isolated more commonly in PWID than in those who do not inject drugs. Other pathogens isolated from infected joints in PWID are similar to those of non-IVDUs, except for higher numbers of viridans streptococci, *Bacillus* spp., *P. aeruginosa*, *Serratia marcescens*, nonfermenting gram-negative rods, nontuberculous mycobacteria, and *Candida* spp. The microbiology of joint infections in PWID has not changed appreciably over the last three decades, except for increases in those due to MRSA.^{21,261–263} Similar to septic arthritis in non-IVDUs, the most common joints involved in septic arthritis in PWID are the knee, hip, shoulder, and wrist.^{21,261–263} Joints that are rarely infected in non-IVDUs that are more commonly septic in PWID include the sternoclavicular, sacroiliac, acromioclavicular, pubic symphysis, and cervical facet joints.^{21,260,264} Other notable clinical features of septic arthritis in PWID include more clinical presentations associated with endocarditis, bacteremia, sepsis, and polymicrobial joint infection.

Management

Initial Approach to the Patient

Patient Assessment

The diagnosis of bacterial arthritis is considered a rheumatologic emergency and should be considered in any patient with acute monoarticular or polyarticular arthritis. Septic arthritis of the hip in a child is an extreme emergency and should be treated in the first 6 to 12 hours.²⁰⁹ Distinct features of the patient's demographic characteristics, clinical history, preexisting joint disease, comorbidity, physical examination, and initial laboratory and imaging may each raise the pretest probability of septic arthritis (see discussion in "Clinical Manifestations" under "Nongonococcal

Arthritis”). A method for estimating the pretest probability of an infected joint has been suggested based on clinical risk factors that are identified in the evaluation of the patient with a painful swollen joint.⁵³ The most useful demographic risk factors are presence or absence of recent joint surgery, RA, advanced age, concurrent skin infection, or diabetes mellitus. The most helpful laboratory parameter is the synovial fluid leukocyte count and differential (see **Table 105.4**). If more than one factor is present, likelihood ratios from each are multiplicative rather than additive. Evidence-based guidelines for the diagnosis and management of septic arthritis have been published and were updated without changes in 2017 by the British Society for Rheumatology Standards, Guidelines and Audit Working Group.²⁶⁵ Appropriateness criteria and suggested algorithms for imaging in septic arthritis were published by the American College of Radiology (ACR) in 2022.²⁶⁶

Laboratory Assessment

As part of the clinical assessment, routine laboratory tests should be drawn for identification of comorbid conditions and as a baseline for future antimicrobial monitoring. ESR, CRP, and serum PCT, although not often predictive for the diagnosis of septic arthritis, may be useful for assessing response to therapy or as a prognostic marker. Serum PCT is helpful only if significantly positive, suggesting bacterial joint infection. Cultures should be obtained from blood, any wound contiguous with the afflicted joint, and skin lesions if present. Plain film joint radiographs should be taken, and if there is concern for soft tissue or osseous extension of infection or prolonged or atypical joint symptoms, CT and MRI should be considered.

Synovial Fluid Assessment

Arthrocentesis should be performed on all patients with inflammatory arthritis in whom joint infection is within the differential diagnosis. Skin visibly involved with cellulitis should be avoided when entering the joint. For cases of polyarticular arthritis, multiple afflicted joints should be aspirated. For axial and deep joints that are difficult to aspirate at the bedside (e.g., sacroiliac or pubic symphysis) or in the event of a “dry tap” of a peripheral joint, fluoroscopic- or CT-guided arthrocentesis is indicated. Synovial

fluid examination should include cell count, leukocyte differential, and crystal examination under polarized light. Tests that need not be performed because of limited utility Page 1342 include viscosity, total protein, lactate dehydrogenase, glucose, lactic acid, and bacterial capsular antigens. Gram stain and aerobic and anaerobic bacterial cultures should be performed. If blood culture bottles or isolator tubes are used, they should be inoculated at the bedside. In patients who are immunocompromised, or in cases with subacute, chronic, or relapsing presentation or penetrating joint trauma, fungal and mycobacterial smear and culture are indicated.²⁶⁷ All cultures should be processed promptly in the laboratory. Synovial fluid multiplex PCR testing, if available, increases the sensitivity for an etiologic diagnosis, particularly for fastidious bacteria, and will shorten the time for initiation of specific directed antimicrobial therapy. Antimicrobial therapy should be delayed until arthrocentesis and appropriate diagnostic cultures are obtained, unless the patient shows signs of sepsis. Orthopedic consultation is often necessary for joint drainage, irrigation, and débridement, and orthopedic or rheumatologic assistance is occasionally needed for diagnostic arthrocentesis.

Other Analysis

If gonococcal arthritis is suspected, NAATs for *N. gonorrhoeae* should be obtained from genitourinary and pharyngeal/rectal sites and freshly voided urine. Although NAATs alone will yield a diagnosis, it is extremely important that culture also be performed in cases that have relapsed or not responded to treatment to allow for antimicrobial sensitivities of an *N. gonorrhoeae* isolate. Because of the high frequency of antimicrobial-resistant gonococcal infections, knowledge of drug susceptibility is vital for guiding therapy in these cases.

Differential Diagnosis

The differential diagnosis of pyogenic arthritis includes several inflammatory joint diseases caused by noninfectious sources, in addition to joint infections caused by nonbacterial pathogens.

Crystalline Arthritis

Acute attacks of the crystalline joint diseases gout and pseudogout are perhaps the greatest mimics of bacterial arthritis. There is

significant overlap in the clinical findings of these two entities because both may occur with monoarticular symptoms of warmth, swelling, pain, and erythema of the involved joint. In addition, systemic signs of inflammation, including fever and elevations of peripheral WBC counts, are often evident. For patients with a history of gouty attacks in stereotypical joints, acute joint inflammation is more likely caused by crystalline than bacterial arthritis; however, in such attacks, arthrocentesis is necessary to rule out concomitant bacterial infection.

The coexistence of crystalline and bacterial arthritis is probably not as rare as previously thought.^{268–271} One retrospective investigation found that of 295 joint synovial fluid aspirates that contained crystals, 5% of the bacterial cultures were positive.²⁷² Of importance, a high synovial WBC count was predictive of concomitant septic arthritis with a sensitivity of 86.4%, specificity of 48.3%, and positive predictive value of only 8.0%. In another study of patients with septic arthritis of the ankle, a large proportion of patients (43.5%) also had crystals found on synovial fluid analysis.²⁶⁸ Because of these findings the presence of crystals in synovial fluid should not be used to rule out bacterial arthritis, and bacterial Gram stain and culture should be performed on all joint aspirates.

Other Causes of Arthritis

RA, systemic lupus erythematosus, spondyloarthropathy, Still disease, rheumatic fever, and Kawasaki syndrome typically occur with subacute polyarticular arthritis associated with specific symptoms and signs of their respective diseases. Although these conditions may be confused with viral arthritides, especially parvovirus B19 or chikungunya arthritis, they are usually easily differentiated from bacterial arthritis. More important, however, is the recognition and diagnosis of bacterial arthritis in a patient with preexisting rheumatoid or other inflammatory arthritis. A sudden increase in inflammation of one or two joints in such patients should raise suspicion of complicating bacterial arthritis, even if it is not accompanied by fever or other systemic evidence of infection, and joint aspiration should be performed.

Reactive arthritis may occur spontaneously or follow a recent gastrointestinal or genitourinary infection, particularly in men.²⁷³ The reactive arthritis may be associated with tenosynovitis

and urethritis, thus mimicking disseminated gonorrhea. Patients with reactive oligoarticular arthritis, however, do not display the characteristic rash of DGI and will often have concomitant sacroiliitis, conjunctivitis, circinate balanitis, or keratoderma blennorrhagica—all conditions absent in DGI.

Hemarthrosis caused by trauma or intraarticular injury can mimic septic arthritis of the knee. In such cases a history of injury is readily apparent. For the rare case of septic arthritis after blunt, nonpenetrating trauma, an aspiration with synovial fluid analysis, and culture is required to establish the diagnosis.

In children, toxic synovitis of the hip mimics bacterial arthritis of this joint. Ultrasound can be useful in the evaluation in addition to synovial fluid evaluation.²⁷⁴

Therapy

Joint Drainage

The management of acute bacterial arthritis requires prompt joint drainage, in addition to antimicrobial therapy. One approach to drainage is through daily closed-needle aspiration of affected joints until there is clinical improvement and the amount and degree of purulence of synovial fluid is minimal. Surgical modalities for drainage include arthroscopy and open arthrotomy, both of which allow visualization and irrigation of the joint, lysis of adhesions, and removal of thick purulent material and inflammatory mediators.^{275–}

²⁷⁹ The choice of drainage procedure may depend on the available clinical resources and should be adapted for the individual patient and involved joint. Iterative needle aspiration is less invasive and likely has similar outcomes if the joint is easily accessible and can be adequately drained.^{275,280,281}

For bacterial infection of the knee, hip, shoulder, wrist, and ankle, drainage by arthroscopy is usually preferred over arthrotomy and may need to be repeated for more severe infections.^{276–287} For these joints an arthroscopic approach has similar or improved outcomes compared with open surgery with fewer complications, less need for a repeat drainage procedure, and shorter hospitalizations. If adequate drainage cannot be obtained by arthroscopy, or in cases of penetrating joint injury, arthrotomy should be performed. Risk factors for failure of a single surgical débridement include a history

of inflammatory arthropathy or diabetes mellitus; involvement of the knee, shoulder, or hip; synovial fluid cell count $>85,000$ cells/mm³; and *S. aureus* as the bacterial isolate.²⁸⁸ Timeliness of debridement after presentation is likely an additional important prognostic indicator. One study showed that joint washout within 24 hours of diagnosis shortened hospital length of stay compared with a >24 -hour delay, but pathogen type and comorbid conditions did not.²⁸⁹ A retrospective study and review of the literature recently questioned this concept and suggested that septic arthritis is not an immediate surgical emergency. However, differences in study design, patient populations, published time delays, involved joints, and reported microbiology limited their conclusions.²⁹⁰ Further data, including more controlled studies, are needed before a delay in surgical drainage can be advocated.

Antimicrobial Therapy

Antimicrobial therapy for native joint bacterial arthritis should be initiated without delay to limit articular destruction, preferably after synovial fluid cultures have been obtained. There are no randomized controlled trials that have evaluated antimicrobial regimens for septic arthritis; thus sound evidence is lacking to guide therapeutic options. A large meta-analysis of antimicrobial therapy for bacterial arthritis did not show an advantage of any one, or combination of, therapeutic agent(s) over another.²⁹¹ Initial antibiotic selection is dictated by likely infecting pathogens as determined by the clinical presentation and synovial fluid Gram stain or molecular testing (**Table 105.5**; also see **Tables 105.2** and **105.3**). Risk factors for infection by an antimicrobial resistant pathogen should be considered.

Initial antimicrobial selection

Vancomycin is indicated as empirical therapy for persons with Gram-positive cocci on a synovial fluid Gram stain or as a component of therapy for those with a negative Gram stain because of the high prevalence of health care–associated MRSA and CA-MRSA joint infection. For patients with Gram-negative rods on Gram stain, an antipseudomonal cephalosporin should be used unless there is a recent or concurrent infection or colonization by an extended-spectrum β -lactamase–producing

pathogen, in which case a carbapenem antibiotic is preferred. Aztreonam or ciprofloxacin are less desirable alternatives due to a relatively high proportion of antimicrobial resistance to these agents in many gram-negative pathogens. Injection or infusion of antibiotics directly into the joint is not recommended.

TABLE 105.5

Recommended Initial Empirical Therapy for Adult Native Joint Bacterial Arthritis Based on Gram stain or Molecular Testing

GRAM STAIN/	PREFERRED ANTIBIOTIC ^a	ALTERNATIVE ANTIBIOTIC ^c
Gram-positive cocci	Vancomycin, 15–20 mg/kg (ABW) daily e 12 h ^b	Daptomycin, 6–8mg/kg daily ^c or linezolid, 600mg every 12 h ^c
Gram-negative cocci^d	Ceftriaxone, 1g every 24h	Cefotaxime, 1g every 8 h
Gram-negative rods^f	Cefepime, 2g every 8 h Ceftazidime, 2 g every 8 h	Carbapenem ^{g,h} (preferred alternative) or fluoroquinolone ^c
Gram-stain negative^f	Vancomycin plus Cefepime or Ceftazidime	Daptomycin ^c or linezolid ^c plus Carbapenem ^{g,h} (preferred alternative) or fluoroquinolone ^c

ABW, Actual body weight; IgE, immunoglobulin E.

^a Unless noted otherwise, dosages are IV for persons with normal renal function.

^b Therapeutic monitoring should target a serum area under the concentration-time curve to minimal inhibitory concentration of 400 to 600 mg-h/L.

^c See text discussion under “Directed Therapy for MRSA Infections.”

^d Equivocal gram-negative morphology should be considered as gram-negative rods.

^e Gram-negative cocci with epidemiology or history suggestive of gonococcal infection should be initially treated per CDC Sexually Transmitted Disease Guidelines using ceftriaxone 1 gm qd.

Alternative therapies have not been suggested for patients with a history of a Stevens-Johnson syndrome or severe IgE-mediated allergy to β -lactam antibiotics. Possible empirical options for high-grade penicillin-allergic patients, pending culture sensitivities, include azithromycin, ciprofloxacin, tobramycin, gentamicin, and spectinomycin (not available in the United States).

† For patients with risk factors for resistant gram-negative pathogens (significant health care exposure, immunosuppression, or recent history of extended-spectrum β -lactamase gram-negative infection or colonization), drug selection should consider the patient's prior bacterial isolates if available, regional or local antibiograms, and population-based resistance patterns.

9 Imipenem, 500 mg every 6 h; *or* meropenem, 1 g every 8 h.

h Usually reserved for patients with penicillin allergy or risk factors for resistant gram-negative pathogens (significant health care exposure, immunosuppression, or history of extended-spectrum β -lactamase gram-negative infection or colonization).

i Ciprofloxacin, 400 mg IV every 8 h or 750 mg PO every 12 h *or* levofloxacin, 750 mg IV or PO every 24 h.

Synovial fluid PCR testing with pathogen specific results, including in many instances identification of antimicrobial resistance genes, can be used in lieu of Gram stain to dictate initial therapy. In such cases earlier targeted therapy is often possible. For patients with a negative Gram stain or PCR, with or without identified crystals, empirical therapy pending culture and/or molecular studies is appropriate even if they have only a modest probability of septic arthritis. In these cases vancomycin plus an antipseudomonal cephalosporin should be used.

For persons with a well-documented or described history of serious allergy to penicillin, in most instances a carbapenem antibiotic can be given. For those with a remote or poorly documented or described penicillin allergy, a cephalosporin antibiotic is often tolerated. Penicillin skin testing, followed by an oral cephalosporin challenge, is useful for determining true immunoglobulin E-mediated allergy and is increasingly available for inpatients in acute care hospitals.

Antimicrobial de-escalation

Empirical treatment should be narrowed or deescalated based on the identity and antimicrobial susceptibility of bacteria identified in synovial fluid, blood, or, in some cases, ancillary cultures or test (e.g., blood culture, or for gonococci: mucosal culture or NAAT). If synovial fluid PCR testing was used to preemptively initiate targeted antimicrobial therapy those results should be confirmed by culture and sensitivity.

For cases in which MSSA is identified, therapy should be narrowed to an antistaphylococcal penicillin (e.g., nafcillin/oxacillin) or first-generation cephalosporin (e.g., cefazolin). Although ceftriaxone is an attractive option for *S. aureus* osteoarticular infections because of once-daily dosing, tolerability, and lower cost, it should be noted that it is intrinsically less active than standard antistaphylococcal agents. There is, however, some clinical evidence to support the use of ceftriaxone for MSSA bone and joint infections. Retrospective studies, including a systematic review of 1640 patients, that compared ceftriaxone to antistaphylococcal beta-lactam drugs for treatment of MSSA infections, including osteoarticular, found noninferiority in terms of 90-day all-cause mortality, hospital readmission, or infection reoccurrence.^{292–294} These studies also confirmed fewer adverse events in the ceftriaxone arm. Conversely, a multicenter, retrospective cohort study comparing these treatments for patients with MSSA primary infections and secondary bacteremia found a higher risk of mortality and treatment failure with ceftriaxone.²⁹⁵ All of these studies have limitations and there remains uncertainty regarding the use of ceftriaxone in MSSA septic arthritis. A multicenter prospective, randomized study is needed.

Directed therapy for MRSA infections

For MRSA joint infections, vancomycin should be continued and therapeutic serum concentration monitoring performed to achieve an area under the concentration-time curve (AUC) to MIC ratio of 400 to 600 mg-h/L.²⁹⁶ Daptomycin is an effective alternative for patients with MRSA native joint infection. Published experience with daptomycin in bone and native joint infection is mostly limited to retrospective, observational, and post hoc studies where comparative efficacy and enhanced safety compared to vancomycin is suggested.^{297–302} However, one nested case-control study; a single

prospective, randomized controlled trial; and a systematic review of daptomycin in osteoarticular infections demonstrated safety and similar efficacy as comparator antibiotics.^{303–305} Although a daptomycin dose of 8 mg/kg/day may be more appropriate for prosthetic joint infection,^{306,307} there are no data to suggest doses higher than 6 mg/kg/day for native joint infection. In recent years daptomycin usage has increased in outpatient parenteral antimicrobial therapy (OPAT) for bone and joint infections that are confirmed or suspected to be due to Gram-positive pathogens. Factors likely responsible for this increased usage include convenience factors, such as once-daily dosing, lack of requirement for serum concentration monitoring and subsequent dose adjustment, and less adverse events, compared with vancomycin.³⁰⁷ Evidence for the efficacy of the oxazolidinones, linezolid or tedizolid, in bacterial arthritis includes observational studies, mostly in prosthetic joint infections and osteomyelitis, and a trial in osteoarticular infections in children.^{308–314} Because linezolid is an oral agent with excellent bioavailability and penetrates bone and joints well, it has appeal for treatment of staphylococcal joint infections. However, adverse drug events, including neuropathy, anemia, thrombocytopenia, and lactic acidosis, have historically limited its use for the prolonged therapy that was often required for these infections.

There is considerable interest in using the long-acting lipoglycopeptides, dalbavancin or oritavancin, for staphylococcal (including MRSA) osteoarticular infections, including native joint septic arthritis.³¹⁵ There are several potential advantages for these drugs, including enhanced adherence and safety especially in treating patients with limited access to health care such as PWID and the homeless. Although expensive, these drugs can be dosed IV once per week, and for MRSA native joint septic arthritis, dalbavancin could be dosed only twice to cover the total duration of therapy.³¹⁵ An increasing number of publications and mounting clinical experience suggest therapeutic efficacy for osteomyelitis and both prosthetic and native joint infection, including in difficult to treat populations.^{316–320} Further study is needed to determine the best dose, dosing interval and duration of therapy for optimal efficacy, safety, and cost-effectiveness.

Ceftaroline, a broad-spectrum cephalosporin with MRSA activity, is potentially suitable for treatment of septic arthritis, but more data are needed before it can be endorsed for routine empiric or directed therapy.^{321–323} Like linezolid, neutropenia is associated with prolonged therapy with ceftaroline.³²²

There is very little evidence to support the use of telavancin or newer tetracycline drugs, including tigecycline, eravacycline, or omadacycline, for the treatment of septic arthritis.³²⁴Page 1344

There have been reports that the performance of vancomycin diminishes for severe infections caused by MRSA that have an isolate MIC ≥ 1 mg/L.^{297,325,326} Further clinical studies are needed to establish whether alternative drugs, AUC/MIC dosing of vancomycin, or use of combination therapy with rifampin enhances treatment for native joint infections caused by these organisms.

Directed therapy for other situations

For patients with a high pretest probability of septic arthritis and negative cultures who are responding to empirical therapy, the initially chosen antimicrobials should be continued to complete a full treatment course. Infections caused by animal or human bites should be treated with antimicrobials that have activity against aerobic and anaerobic oral flora—for example, ampicillin-sulbactam or amoxicillin-clavulanate—and in the penicillin-allergic patient, clindamycin plus ciprofloxacin or moxifloxacin.

For infections due to Gram-negative bacteria resistant to antipseudomonal cephalosporins as well as carbapenems and fluoroquinolones, there are few data to guide therapy. Beta-lactam/beta-lactamase inhibitor combination drugs such as meropenem/vaborbactam and ceftazadime/avibactam or other novel agents such as cefiderocol potentially have activity against these organisms but only limited observations or studies have been performed.³²³

Gonococcal arthritis is treated with ceftriaxone, 1 g IM or IV once every 24 hours. After improvement, oral “step-down” therapy may be substituted if culture and sensitivity results are available.²⁴¹

Duration of therapy and use of oral antibiotics

The duration of therapy for native joint septic arthritis remains unclear. Although therapy should be long enough to eradicate all foci of infection and thus prevent recurrence and further joint

disability, the duration does not have to be as long as what is usually given for osteomyelitis or prosthetic joint infection. Potential variables that likely influence the duration of therapy include the particular joint involved, associated bacteremia, identity of the infecting pathogen, adequacy of surgical drainage, and host comorbidity and immune function.

Traditionally, parenteral therapy for 2 to 3 weeks is prescribed, with early conversion to oral therapy if fluoroquinolones or linezolid (or tedizolid) are used. For *S. aureus* (including MRSA) at least 2 weeks of parenteral therapy (oral is acceptable if linezolid/tedizolid) should be given followed by oral therapy to complete 3 to 4 weeks.

Choices for IV to PO switch for *S. aureus* after adequate surgical drainage and 2 weeks of parenteral therapy include linezolid/tedizolid, trimethoprim-sulfamethoxazole (TMP-SMX), doxycycline or minocycline, or clindamycin, depending on susceptibility.

Infections caused by *S. aureus*, including MRSA, with associated bacteremia, but not endocarditis or disseminated infection, can be treated for 4 weeks. Disseminated gonorrhea or gonococcal arthritis requires 1 week of treatment.

There are accumulating data to support shorter courses of antibiotics and much earlier conversion from parenteral to IV therapy for septic arthritis. A prospective, randomized, noninferiority trial of 154 cases of native joint bacterial arthritis showed that 2 weeks was noninferior to 4 weeks of antibiotic therapy.³²⁷ In this trial, over half of the patients had infection of the hand and/or wrist, surgical arthrotomy was performed for drainage in all but 7 patients, few patients had drug-resistant organisms, and the majority of patient infections were due to organisms other than *S. aureus*. Intravenous therapy was given for only 1 to 2 days followed by oral therapy in both study arms. In addition, a retrospective investigation of adult patients showed that rates of recurrent infection and sequelae of infectious arthritis did not differ among patients who received 7 days of parenteral antibiotics, 8 to 21 days, or greater than 21 days, as long as successful drainage was performed.³²⁸ In the same study, patients who had 14 days or less of total antibiotic treatment, which included initial parenteral therapy followed by oral therapy, had no difference in recurrence rates or sequelae compared with those who received longer treatment

courses. Reported risk factors for recurrence of infection included immune suppression and the lack of surgical intervention. A second retrospective study of 14 days of antimicrobial therapy for pyogenic arthritis of the fingers and wrist reported good outcomes if surgical débridement was adequate.³²⁹

Therefore it is reasonable to assume that a short (1–2 days) duration of parenteral followed by oral therapy to complete 2 weeks is likely appropriate for immunocompetent patients with infections of small joints due to easy to treat bacterial pathogens. However, for infections due to *S. aureus* or other difficult to treat pathogens (i.e., *P. aeruginosa*) or of large joints or in patients with comorbidities, longer courses should be given. Further clinical trials to conclusively answer the question of optimal duration of therapy are needed.

Adjuvant Therapy

Current research is examining whether adjuvant therapy directed at the attenuation of microbial virulence factors or the host inflammatory response is able to reduce the sequelae of bacterial joint infection.^{24,55} There is some evidence that adjuvant early, short-course, systemic corticosteroid treatment for children with hematogenous bacterial arthritis significantly reduces the duration of symptoms, hospitalization, antibiotic therapy, and residual joint dysfunction.^{330–333} A 2018 Cochrane Review could draw no strong conclusions from these and other studies other than that pain and functional outcome might be improved and duration of antimicrobial therapy possibly reduced. They recommended further randomized studies with relevant outcomes.³³⁴

Animal studies have examined TNF- α antagonists, IL-12, recombinant IL-10, bisphosphonates, and other modulators of inflammation as adjuvant treatments with some benefits.^{26,163,24,161,335} Although these approaches look promising, further research is needed.

Viral Arthritis

Arthritis or arthropathy caused by viral agents is often acute, occurs concurrently with signs and symptoms of febrile systemic illness, and resolves along with other manifestations of illness. Viruses may cause arthritis directly by infecting the synovium or indirectly

through host immune-mediated responses, and much of the pathogenesis is poorly understood.^{336,337}

Parvovirus B19

Perhaps the most common form of viral arthritis in developed countries is caused by human parvovirus B19. In children human parvovirus B19 infection arises with fever, rash, coryza, headache, and malaise.³³⁸ The rash classically arises with bright red “slapped cheeks” and a lacy, reticular exanthem on the torso and extremities. Arthritis occurs in less than 5% of afflicted children and tends to be asymmetrical and pauciarticular, and it predominantly affects the knee.^{339,340}

Adults who become infected with human parvovirus B19 do not display the classic facial erythema noted in children and present with a febrile, flulike illness with prominent polyarthralgia and joint inflammation accompanied by an absence of reticulocytosis.^{336,339–341} Cases are usually seen in late winter and early spring, and regional and nosocomial outbreaks of disease have been reported.^{342–345} An associated exanthem, if apparent, is a fleeting and difficult-to-appreciate rash on the torso and, less often, on the extremities, which may show some of the reticular aspects exhibited by children (Fig. 105.5). Arthritis or prominent arthralgias are more common in females than men with human parvovirus B19 infection, affecting 60% and 30% of individuals, respectively.^{339–346} Characteristically, an acute and moderately severe, symmetrical polyarthritis of the proximal interphalangeal and metacarpophalangeal joints is associated with morning stiffness and somewhat resembles RA.^{338,339,347} Knees, wrists, and ankles may also be involved. In approximately one-third of patients, particularly women, joint symptoms occur in episodic flares with symptom-free intervals.³³⁸ A biphasic illness may manifest in some patients with an acute febrile illness that is separated from rash and joint symptoms by several days.³⁴⁸ Diagnosis is made clinically, and a positive serologic or peripheral blood PCR test supports the diagnosis, although false-positive and -negative results occur with both tests.^{336,337,349}

Although joint symptoms usually resolve within 2 weeks, persistent polyarticular arthritis may follow acute human parvovirus B19 infection in up to 20% of female patients and, in some individuals, may last up to several months, occasionally longer.^{336,338,350} In such patients joint erosions and objective evidence of synovial inflammation do not develop,³³⁸ and follow-up studies do not demonstrate long-term joint swelling or restricted range of motion consistent with true chronic arthritis.³⁵¹ Treatment of chronic human parvovirus B19 arthritis is supportive with nonsteroidal anti-inflammatory drugs (NSAIDs), and, except in exceptional cases, IV immunoglobulin is not indicated.^{336,352} Page 1345 The role of human parvovirus B19 in the pathogenesis of RA or systemic lupus erythematosus remains circumstantial and inconclusive.^{337,339,353}

FIG. 105.5 Photographic image of the faint, lacy reticular rash of the proximal lower extremities characteristic of adult human parvovirus B19 infection.

Follow for extended description

Alphaviruses, Including Chikungunya

Several alphavirus infections are associated with arthritis and are distributed worldwide, particularly in tropical and subtropical regions.^{337,354,355} These arboviruses usually cause sporadic and endemic disease; however, viral mutation or an increased prevalence of their mosquito vectors may result in periodic explosive outbreaks of fever, rash, and arthritis.^{355–357} In 2005 an outbreak of chikungunya virus infection emerged in the islands of the Indian Ocean, subsequently spreading across global tropical and subtropical regions, eventually peaking in 2016 after millions of cases.^{355–359} These included several imported and a handful of locally acquired cases in Europe and the continental United States.³⁶⁰ Since 2016 the virus has remained endemic throughout the tropics and subtropics worldwide and still sporadically causes focal or regional epidemics.^{357,361}

Chikungunya virus is transmitted by the bite of the daytime-biting *Aedes* spp. mosquitoes. The presence of the mosquito vector, *Aedes*

albopictus, in Europe and the Americas, together with returning travelers with extremely high levels of viremia, is a public health concern in that infection could be easily established in these previously nonendemic subtropical regions.^{355,359,362}

Chikungunya fever is characterized by an acute biphasic febrile illness, with the initial phase manifesting headache, malaise, myalgias, symmetrical arthralgias, or polyarthritis and in up to 70% of cases, a skin rash (Fig. 105.6).^{354–359,361,362} Joint symptoms are often quite severe (chikungunya means “that which contorts or bends up”), affecting predominantly the wrist, ankles, and small joints of the hands and feet. After 5 to 7 days of fever and a brief interlude of improved symptoms, a polyarthritis with or without tenosynovitis often recurs and may be disabling.^{355,357,361–363} Some cases may resemble dengue fever or Zika virus infection, both of which are also highly endemic in regions where chikungunya fever occurs. Similar to infection caused by human parvovirus B19, a chronic form of chikungunya virus–associated arthritis has also been described.^{364,365} A meta-analysis including 6532 patients with chikungunya virus arthritis showed that 43% did not fully recover after 3 months, and 21% were still symptomatic within 1 year. In many patients symptoms persisted for more than 28 months.³⁶⁴ Up to one-fourth of those afflicted rated their symptoms as severe. Several studies show that older age, female sex, diabetes mellitus, osteoarthritis, and more prominent symptoms including symmetrical joint involvement during the acute phase were associated with persistence of joint symptoms.³⁶⁶ In addition, the level of acute chikungunya viremia may also be a factor.^{366,367} A different experience has been reported from a rheumatology clinic in the Dominican Republic, where the mean duration of symptoms was 2.5 months, and 89% of 457 patients presenting with chikungunya virus infection responded well to NSAIDs, although 72% were treated with low-dose glucocorticoids (e.g., 5–7.5 mg of prednisone daily).³⁶⁸ Patients with RA, all of whom had been on and synthetic and biologic DMARDs, did equally well.

FIG. 105.6 Diagnostic image of the erythematous maculopapular rash with islands of normal skin characteristic of chikungunya virus infection. From Hochedez P, Jaureguiberry S, Debruyne M, et al. Chikungunya infection in travelers. *Emerg Infect Dis*. 2006;12:1565–1567.

Follow for extended description

Diagnosis of chikungunya fever–associated arthritis is based on clinical, epidemiologic, and laboratory criteria. Tests of acute and convalescent serum with PCR and indirect immunofluorescence and enzyme-linked immunosorbent assay serology are most useful.^{355–358,361,362} Recommended treatments of chronic chikungunya arthritis, all with a low level of evidence, include NSAIDs, low-dose corticosteroids, and DMARDs.³⁶⁹

Other arboviruses that cause alphaviral arthritis include O'nyong-nyong, Igbo-Ora, Ross River, Barmah Forest, Sindbis, and Mayaro viruses.³⁵⁴

SARS-CoV-2

COVID-19, caused by SARS-CoV-2, is a novel respiratory infection first described in Wuhan, China, in December 2019 responsible for a pandemic that resulted in millions of deaths and significant socioeconomic impact across the world (see Chapters 132–136). Although COVID-19 is largely an upper and lower respiratory infection, several extra–respiratory system complications are responsible for much of the morbidity and mortality of more severe infection. Although still relatively unclear, many of these complications, including in the musculoskeletal system, are likely a result of the associated inflammatory response and cytokine storm rather than from direct viral infection.³⁷⁰ Most patients recover from SARS-CoV-2 infection without significant joint manifestations, even though approximately 15% will note arthralgia. Postinfectious inflammatory arthritis, however, has been well described following COVID-19, occasionally associated with new-onset autoimmune and vasculitic diseases.³⁷¹ Molecular mimicry is postulated to be involved in the pathogenesis.

Although relatively uncommon, reactive arthritis is the best described COVID-19–associated inflammatory arthritis, usually occurring 2 to 4 weeks after infection.³⁷² It mostly presents with classic oligoarticular reactive arthritis of the lower extremity; however, axial spondyloarthritis and a more rheumatoid arthritis-like presentation have also been described. Unlike classical reactive arthritis, cases seem to be equally distributed between men and women and affect those older than 50 years of age. Fortunately, the majority of patients respond to NSAIDs and arthritis resolves within a few weeks. Those with more rheumatoid arthritis presentations may have a more chronic course. Although the causal relationship is unclear, a small number of cases of reactive arthritis have also been described following COVID-19 vaccination.^{372,373}

As COVID-19 infection continues to evolve in the postpandemic period due to population immunity and emergence of new variants with different virulence properties, the incidence and clinical aspects of post-COVID reactive arthritis will likely also evolve. Page 1346

Other Causes of Viral Arthritis

Rubella is another important cause of viral arthritis in adults, particularly in the developing world, where childhood vaccination is not widespread. Joint inflammation may follow natural infection or immunization with live-attenuated vaccine.^{336,374} Arthritis is thought to occur via direct infection of joint synovial tissue, although immune complexes may also play a role.^{337,375,376} As with human parvovirus B19, the polyarthritis that is associated with rubella affects predominantly adolescent girls and women, most frequently involving the small joints of the hands and, less commonly, the knees, wrists, and ankles.^{337,375,377} Arthralgia and arthritis occurred simultaneously or within a few days of the rash in 18% and 21% of adult male and female subjects, respectively, in a 2012–13 outbreak of wild-type virus rubella infection in Tokyo, Japan.³⁷⁸ Arthropathy can also occur in seronegative individuals following rubella vaccination. One study found joint symptoms, including arthritis, in 14% of adults 2 to 3 weeks after receiving rubella vaccine.³⁷⁷ Furthermore, a placebo-controlled, randomized

trial showed that 30% of seronegative adult women receiving rubella vaccine developed acute arthropathy after vaccination, as opposed to 20% of those receiving placebo, a significant difference.³⁷⁹ Arthritis after rubella vaccination or natural infection may persist for several weeks, but chronic or recurring arthritis is unusual.^{374,376,380} Treatment is supportive, with NSAIDs or low to moderate doses of corticosteroids.³³⁶ Of note, MMR booster vaccine has been found to be safe and to not exacerbate disease activity in children with juvenile idiopathic arthritis.³⁸¹

In acute hepatitis B infections severe arthralgia or arthritis may be seen during the febrile prodrome before the onset of jaundice and is often seen together with urticaria.^{336,382,383} A symmetrical, polyarticular arthritis typically involves the hands, wrists, knees, or ankles and resolves with the onset of jaundice. The clinical findings may resemble an RA syndrome in the acute infection period, and a positive rheumatoid factor may be seen in 40% of patients.³⁸² Although an association between chronic hepatitis B virus (HBV) infection and RA has been hypothesized, sound evidence for this is lacking despite RA patients having a greater proportion of chronic HBV than matched controls.^{382,384,385} Patients with rheumatoid arthritis, though, should be screened for chronic HBV, particularly before DMARD therapy.³⁸⁵

Arthritis in the setting of hepatitis C virus (HCV) infection is poorly described. Joint symptoms are noted in up to 50% of patients with HCV infection, although only 2% to 4% have frank arthritis.^{336,382} It is controversial whether HCV infection is associated with a distinct inflammatory arthritis syndrome, although there is growing evidence that a rare nonerosive inflammatory arthritis mimicking RA may occur.^{336,382} In chronic hepatitis B– or hepatitis C–induced essential mixed cryoglobulinemia, arthralgias and fibromyalgia are described as occurring in approximately one-third of patients, although frank arthritis is not usually observed.^{382,386,387} Of interest, there is some evidence that direct-acting antivirals against hepatitis C improve RA activity and treatment outcomes in patients with HCV, adding some evidence to the association between the two entities.³⁸⁸

Both HIV-1 and human T-cell leukemia virus 1 (HTLV-1) are retroviruses with the ability to cause arthritis or arthropathy.^{336,389} Nonspecific joint pain is common in the setting of HIV infection, although true articular manifestations of HIV-1 are rare and include chronic arthralgia and oligoarticular arthritis.^{336,390,391} An increased incidence of reactive and psoriatic arthritis has been observed in patients with HIV, particularly among MSM.^{336,390} Some elderly patients in Japan infected with HTLV-1 have been described with a chronic joint infection termed HTLV-1–associated arthropathy, which closely resembles RA (see Chapter 175).^{389,392}

Musculoskeletal complaints in survivors of Ebola virus infection are common, with 70% complaining of myalgia and arthralgia 3 weeks after hospital discharge.^{393,394} Longer-term joint complications in these patients may include symmetrical polyarticular arthralgia and arthritis, enthesitis, and tendon rupture. Overt joint inflammation is usually not apparent.

Chronic Infectious Arthritis

Chronic infectious arthritis consists of a constellation of monoarticular or, less commonly, oligoarticular joint infections that are characterized by an insidious onset and indolent course, a paucity of symptoms, and progressive joint destruction that may result in considerable loss of articular function. Many of these infections arise with few symptoms and signs of joint inflammation other than subacute or chronic joint swelling, with or without effusion, and pain and stiffness during active range of motion. Although this form of infectious arthritis remains relatively uncommon compared with the incidence of acute septic arthritis, it is a significant component of the differential diagnosis of subacute or chronic joint inflammation. An increasing number of anecdotal reports indicate that it is an emerging problem in immunocompromised or chronically ill hosts. Noteworthy aspects of this entity include its ability to mimic other inflammatory joint disorders, such as RA, and its ability to arouse little clinical suspicion, resulting in considerable delays in diagnosis. Moreover,

establishing a pathogen-specific diagnosis is difficult, and the response to treatment is slow and often incomplete. Inappropriate treatment—for example, systemic or intraarticular steroids for a tentative diagnosis of RA—is not uncommon and can further delay diagnosis and/or lead to more rapid or severe joint destruction. Subacute or chronic infectious arthritis is usually caused by mycobacteria or fungi, and occasionally by bacteria, such as *B. burgdorferi*, *T. whipplei*, *Treponema pallidum* (tertiary syphilis),³⁹⁵ and *Nocardia* (Table 105.6).³⁹⁶ Chronic arthritis caused by parasitic infection has been rarely described with some helminth infections such as lymphatic filariasis.^{397,398}

Fungal Arthritis

Fungal arthritis may sometimes occur in normal hosts; however, in immunocompromised or chronically ill persons, a steadily increasing frequency of infection and diversity of infecting mycotic pathogens is apparent.^{399,400} Although there is some overlap between the two groups, the most common fungal pathogens isolated from infectious arthritis in healthy hosts residing in endemic regions for the dimorphic fungi are *Blastomyces dermatitidis*, *Coccidioides* spp., and *Sporothrix* spp., whereas in immunocompromised hosts *Candida* spp., *Cryptococcus*, and *Aspergillus* or other molds are more often observed.^{399–401} However, pulmonary and disseminated infections caused by the dimorphic fungi are increasingly diagnosed in patients with RA or IBD on TNF- α inhibitor therapy, and these drugs are likely a risk factor for chronic fungal arthritis.^{401–403} Joint infection from fungal pathogens usually results from hematogenous dissemination of the organism and, except for *Candida*, has as its source a symptomatic or subclinical respiratory tract infection. On occasion, a fungus, usually a mold, is introduced into the joint via direct trauma or injury, sometimes associated with a penetrating foreign body.^{400,404}

Candida Arthritis

Candida arthritis afflicts both native and prosthetic joints and has been increasing in frequency over the last two decades, parallel to the rising incidence of systemic and focal infections caused by this

organism in hospitalized and chronically ill patients.^{401,405–408} Risk factors for joint infection with this organism include loss of skin integrity, diabetes mellitus, hemodialysis, malignancy, surgery, malnutrition, premature birth, IVDU, immunosuppressive therapy (including corticosteroids), and the prolonged use of broad-spectrum antimicrobials, hyperalimentation fluid, and central IV catheters.^{21,399,401,406–408} In most cases articular infection occurs through the hematogenous route, with normal joints typically afflicted in a monoarticular pattern, although polyarticular infection occurs in up to 30% of cases.⁴⁰⁷ In one retrospective study of 765 patients with documented candidemia, 0.4% were found to have subsequent *Candida* osteoarticular infection.⁴⁰⁹ Infection can also occur by direct inoculation of the joint through surgery or corticosteroid injection, particularly in arthritic or surgically altered joints.^{399,407} Patients are often heavily colonized by *Candida* at skin, gastrointestinal, or mucosal sites before joint infection. *Candida albicans* is the most commonly isolated species,^{401,406,407} followed by *C. tropicalis* and *C. parapsilosis*. Other less commonly cultured species include *C. glabrata*, *C. krusei*, *C. guilliermondii*, *C. zeylanoides*, *C. lambica*, and *C. lusitaniae*.^{399,406,410–415} Osteoarticular infections due to the extensively-drug resistant *C. auris* have rarely been reported.⁴¹⁶

As in bacterial arthritis, the knee is affected in the vast majority of cases and less often the hip or other joints.^{401,406,407} In IV drug abusers, fibrocartilaginous joints, including the sternoclavicular, sacroiliac, and Page 1347costochondral, may be involved. In addition, they may manifest with associated *Candida* endophthalmitis and folliculitis of the face, scalp, and torso.⁴¹⁷

TABLE 105.6

Infectious Causes of Chronic Monoarticular or Oligoarticular Arthritis

Bacterial

- *Borrelia burgdorferi*
- *Tropheryma whippelii*

- *Treponema pallidum*
- *Nocardia* spp.

Fungi

- *Candida* spp.
- *Cryptococcus neoformans*
- *Blastomyces dermatitidis*
- *Coccidioides* spp.
- *Paracoccidioides brasiliensis*
- *Sporothrix schenckii*
- *Aspergillus* spp. and other molds, including *Rhizopus*, *Scedosporium*, *Fusarium*, *Lomentospora*, and *Mucorales*

Mycobacteria

- *Mycobacterium tuberculosis*
- *M. kansasii*
- *M. marinum*
- *M. avium* complex
- *M. terrae*
- *M. fortuitum*, *M. chelonae*, *M. abscessus*
- *M. haemophilum*
- *M. leprae*

Parasites

- Helminths
- Filariae

Candida arthritis may arise acutely, with reports of marked joint inflammation and associated fever and constitutional symptoms, especially in cases accompanied by systemic candidiasis or, more insidiously, with chronic indolent joint pain and stiffness and minimal systemic manifestations.^{407,409} In the latter case a long latent period is often seen between an episode of apparent or suspected occult candidemia and the onset of joint complaints. Moreover, *Candida* arthritis may emerge during the course of appropriate therapy for candidemia.⁴⁰¹ Extension of infection from the joint to adjacent bone is not uncommon.³⁹⁹ Synovial fluid analysis characteristically shows polymorphonuclear leukocytosis with cell counts greater than 25,000 cells/mm³ and a normal or low measured glucose. Gram stain of synovial fluid is not usually diagnostic, although bacterial or fungal culture yields *Candida* spp.

in most cases. Blood cultures are positive in less than 30% to 50% of patients.

Cryptococcal Arthritis

Cryptococcus neoformans causes skeletal infections in less than 10% of patients with disseminated infection with this organism.^{418–420} Cryptococcal arthritis is considerably less common than cryptococcal osteomyelitis, and the latter is frequently the nidus of infection for the former through local contiguous spread. The majority of patients with joint infection caused by this organism are immunocompromised because of various conditions, including acquired immunodeficiency syndrome.^{419–421} The presentation of cryptococcal arthritis and associated osteomyelitis is that of a chronic joint infection, most often involving the knee.^{399,419,420} A serum cryptococcal antigen test is often positive.⁴¹⁹ Diagnosis requires joint aspiration or synovial biopsy with appropriate synovial fluid or tissue fungal cultures. Of importance, with rare exceptions, osteoarticular infection represents disseminated cryptococcal disease; thus coexisting infection of the central nervous system should be ruled out.

Arthritis Due to Dimorphic Fungi

Acute infection caused by *Coccidioides immitis* or *Coccidioides posadasii* occurs in the Southwestern region of the United States, northern Mexico, and a few regions of Central and South America, causing a symptomatic or subclinical primary pneumonitis.

Migratory polyarthralgia and polyarticular arthritis occur in approximately one-third of these patients, manifesting as a self-limited hypersensitivity syndrome termed “desert rheumatism” or “valley fever.”⁴⁰⁰ In about 1% to 7% of persons with coccidioidomycosis, dissemination occurs, and although any host may be afflicted, it is more likely in men, pregnant females, Blacks, Filipinos, and immunocompromised persons, including patients with HIV infection.^{422,423} Infections in children are well described.⁴⁰¹ Skeletal involvement occurs in approximately 30% of cases and results in a chronic granulomatous infection of bones, joints, and periarticular tissues.^{399,401,422,423} Joint infection may be primary or secondary to extension from contiguous

osteomyelitis. Weight-bearing joints, particularly the knees and ankles, are most frequently affected, but any joint may be involved.^{401,422,424}

Presentation is that of subacute or chronic monoarticular or polyarticular arthritis with the most common symptom of bone or joint pain. The infection may progress either slowly or quite rapidly, resulting in extensive joint destruction.^{401,422,425} Other manifestations of disseminated infection are not uncommon at the initial evaluation.⁴²² Up to one-third of cases have multiple bone and joint lesions at diagnosis.⁴⁰¹ Peripheral eosinophilia may be present, and synovial fluid leukocyte counts are moderately elevated, with a lymphocytic predominance. Coccidioidal arthritis should be considered in patients presenting with chronic progressive arthritis in endemic areas or in those who have traveled to them. Diagnosis is suggested by an elevated titer against coccidioidal antigens on serologic testing of serum or visualization of spherules in synovial fluid. Confirmation is by fungal culture of synovial fluid and/or histopathologic examination and culture of synovial or periarticular tissues.⁴²⁶ Synovial fluid PCR testing may have utility for diagnosis.⁴⁰¹ In suspected cases of coccidioidal arthritis the microbiology laboratory should be alerted before sending specimens for culture because *Coccidioides* spp. are important biohazards in the clinical laboratory. Once isolated, shipping the isolate for confirmation of identity is complicated because the organism is a potential agent of bioterrorism, and its transport is strictly regulated.⁴²⁷

Similar to coccidioidomycosis, blastomycosis is primarily a pulmonary infection acquired through inhalation that may later disseminate, potentially involving the bones and joints. Skeletal infection caused by *Blastomyces dermatitidis* occurs in about one-fourth of patients with disseminated infection, and joint infection in most cases is associated with juxtaarticular osteomyelitis.^{401,428} Also similar to coccidiomycosis, approximately 75% of bone and joint infections occur in men.⁴⁰¹ Monoarticular arthritis of the knee is most common, followed by the ankle, elbow, wrist, and hand.³⁹⁹ Polyarticular infection is extremely rare.^{429,430} An acute arthritis

with abrupt onset and accompanying systemic symptoms of inflammation and a more chronic joint infection are both described.^{401,429} Concomitant blastomycosis pneumonia and nodular or ulcerative skin lesions are frequently evident on examination, and periarticular draining sinus tracts are occasionally seen.⁴⁰¹ Synovial fluid is purulent, and organisms are readily demonstrated on cytologic examination or microscopic examination of a wet mount smear of draining pus or synovial fluid.⁴⁰¹ Cultures of synovial fluid or other affected locations will usually confirm the diagnosis but should be incubated for at least 4 weeks.

Histoplasma capsulatum is an extremely rare cause of fungal arthritis in endemic areas, usually seen in adults rather than in children, and is usually associated with immunosuppression including corticosteroids and DMARDs.^{331,431} A chronic monoarticular arthritis of the knee is most common, but also hip, wrist, ankle, and shoulder infections are seen.^{401,431,432} Tuberculosis arthritis presents with similar clinical and histopathologic manifestations and is in the differential diagnosis. Synovial fluid or tissue stain, culture, and/or histopathology will usually yield the diagnosis.⁴⁰¹ In addition to direct joint infection, a hypersensitivity syndrome with rash and arthralgia may accompany *Histoplasma* primary infection, which is similar to that in coccidioidomycosis.^{433,434}

Sporotrichosis is a chronic infection caused by *Sporothrix* spp. that primarily affects the skin and secondarily the lymphatics after percutaneous inoculation by a contaminated foreign body that is carrying plant or soil material (see Chapter 265 for new *Sporothrix* spp. names). In contrast to the other endemic mycoses, primary pulmonary infection is uncommon. Although rare, osteoarticular sporotrichosis comprises approximately 80% of extracutaneous infections and results from hematogenous dissemination.⁴⁰¹ On occasion, hematogenous spread to the skin occurs at the same time. The portal of entry is usually Page 1348not apparent, although a pulmonary portal is sometimes evident. At most, only 0.3% of patients with cutaneous sporotrichosis will develop

arthritis.^{399,435} Unlike cutaneous sporotrichosis, infection from direct joint inoculation by a thorn or plant material is rare.^{401,404}

Immunocompromised patients, patients with diabetes, and patients with alcoholism are at higher risk for disseminated *Sporothrix* spp. infection and arthritis,^{434,436–438} and the typical patient is a male involved in gardening, farming, or an outdoor occupation.^{401,439} Zoonotic infections from contact with cats, and less commonly dogs, has been reported.⁴⁰¹ Chronic monoarticular or polyarticular sporotrichosis arthritis arises in the knees, wrists, hands, elbow, and ankle, and a tenosynovitis has also been reported.^{399,439,440} Symptoms and signs are more prominent in the small joints of the wrist and ankle. Fever is uncommon, and there are usually no other focal symptoms or signs of infection, including pulmonary or cutaneous, before or during the course of arthritis.^{438,439} Diagnosis is difficult because organisms are rarely found on histologic examination of joint fluid or tissue, and fungal culture of these specimens is often negative. Multiple aspirations and biopsy of affected joints may be necessary. As a result, extremely long diagnostic delays and permanent joint sequelae are common.⁴⁴¹

Arthritis Due to Molds

Molds, including *Aspergillus*, are an exceedingly rare cause of septic arthritis, with the majority of cases having been reported in association with disseminated infection in immunocompromised hosts, including chronic granulomatous disease, or introduction by trauma.^{442–444} One review described 31 cases of *Aspergillus* septic arthritis and noted the knee and hips to be the most frequently infected joints.⁴⁴⁵ Many patients had oligoarticular involvement due to hematogenous seeding. *Aspergillus fumigatus* is the most common isolate seen in approximately 50% to 75% of cases, followed by *A. flavus* in 15% and *A. niger*, *A. terreus*, or *A. niger* in 3% to 10% each.^{442–444} Osteolysis and tissue extension are common. Synovial fluid WBC count varies considerably, with a median of 17,200 cells/mm³.⁴⁴³ Diagnosis is established through cultures or histopathology. Prognosis is poor, with an overall survival of 65–85% depending on comorbidities.^{442,443}

Fungal arthritis caused by *Fusarium* spp., *Curvularia* spp., *Exophiala* spp., *Trichosporon asahii*, *Lomentospora* (*Scedosporium*) *prolificans*, *Scedosporium apiospermum* (*Pseudallescheria boydii*), and *Mucorales* have all been described and may be seen in either immunocompromised or immunocompetent patients.^{401,445–450} A review of 43 years of reports of non-*Aspergillus* mold osteoarticular infections found antecedent trauma in children and prior surgery in adults as risk factors.⁴⁵¹ Hyaline molds were the most common agents reported, among which *S. apiospermum* and *L. prolificans* were most frequent. All of these pathogens usually cause severe joint infections and are difficult to diagnose, and many respond poorly to surgery and antifungal therapy.

Bone and joint infections caused by *Aspergillus* spp. and *Rhizopus* spp. have complicated arthroscopic anterior cruciate ligament reconstruction in young healthy persons.^{452,453} Mold infections after intraarticular steroid injections in immunocompetent hosts have also been reported.^{442,454} In the fall of 2012 a multistate outbreak of fungal infection associated with the injection of contaminated methylprednisolone acetate from a single compounding pharmacy resulted in several cases of fungal meningitis, vertebral osteomyelitis, epidural abscess, and peripheral joint infection. Altogether, there were 33 cases of peripheral joint infection caused by *Exserohilum* spp., some of which presented several months after joint injection with contaminated steroid.^{35,156,455} Thirty-seven percent of patients showed clinical cure and 63% showed improvement at 12 months after 6 to 12 months of azole therapy.⁴⁵⁶

Therapy for Fungal Arthritis

A combined approach to therapy for fungal native joint arthritis, using medical and surgical modalities, is typically necessary for optimal results. Open or arthroscopic débridement and drainage should be performed, at which time tissue and fluid specimens can be obtained for histopathologic and microbiologic testing. However, when a diagnosis of cryptococcal, *Sporothrix*, or perhaps

coccidioidal joint infection has already been established, drainage and debridement are often not necessary.^{401,457,458}

Historically, most of the clinical experience with therapy for fungal arthritis was with amphotericin B, with or without the addition of flucytosine. In the past three decades fluconazole, newer extended-spectrum triazoles (itraconazole, voriconazole, posaconazole, and isavuconazole), terbinafine, and the echinocandin class antifungals (caspofungin, micafungin, and anidulafungin) have expanded the therapeutic options available for many fungal pathogens.^{459–464} In addition, lipid preparations of amphotericin B have modestly reduced the toxicity of this drug, increasing the safety of long-term amphotericin therapy.

Treatment guidelines for native joint *Candida* arthritis are based on case reports and open-label series. Antifungal susceptibility testing should be performed on all culture isolates. Recommendations for fluconazole-susceptible isolates include oral fluconazole, 400 mg (6 mg/kg) daily for at least 6 weeks, or an IV echinocandin daily for at least 2 weeks, followed by oral fluconazole to complete therapy.⁴⁶⁵ Intravenous fluconazole should be used for patients intolerant of oral treatment. Alternatively, a lipid formulation of amphotericin B, 3 to 5 mg/kg/day, can be used in place of the echinocandin, although the risk of nephrotoxicity makes this choice less appealing. Intraarticular amphotericin B is not recommended. Courses of therapy longer than 6 weeks are frequently necessary. Fluconazole, if used, may be given orally.

Echinocandins are the preferred therapy for *C. glabrata* and *C. krusei* joint infections due to relative or absolute resistance to fluconazole. For *C. glabrata*, if testing shows susceptibility to fluconazole, therapy can be switched to this oral drug after 2 weeks of treatment. Oral voriconazole can be considered for step-down therapy for septic arthritis due to *C. krusei*, as most isolates are susceptible to this drug.^{465–467} The incidence of *Candida* resistance to echinocandins is quite low, whereas the proportion demonstrating azole resistance depends on the species.^{466,468} For *C. parapsilosis* increased minimum inhibitory concentration results

are often seen. For these patients 6 to 8 weeks of fluconazole, rather than echinocandins, is preferred. Surgical débridement, sometimes multiple procedures, is indicated in all cases of Candida arthritis.⁴⁶⁵ Successful eradication of Candida joint infection is best confirmed by posttreatment synovial fluid analysis and culture.

For cryptococcal arthritis the choice of treatment depends on the extent of disseminated disease and the immune status of the patient. In general, initial therapy with an amphotericin B preparation plus/minus flucytosine, followed by oral fluconazole, is appropriate, but in nonimmunocompromised patients with isolated septic arthritis, treatment with fluconazole, 400 mg daily for 6 to 12 months, may be appropriate.^{401,457} Some treatment courses may need to be extended beyond 12 months. Arthrotomy or arthroscopic débridement is seldom necessary.

The preferred treatment of isolated joint infection without central nervous system involvement caused by the endemic mycoses, including *Coccidioides*, *Blastomyces*, *Histoplasma*, and *Sporothrix*, is with itraconazole, 200 mg twice daily for at least 12 months, unless the patient has limb-threatening skeletal disease, in which case an induction course of liposomal amphotericin B preparation is advised.^{458,469–471} Measurement of serum itraconazole levels is suggested during therapy. Coccidioidal joint infection can also be treated with high doses of fluconazole (400–800 mg daily). With either drug, a longer duration of therapy may be necessary.⁴⁷⁰ Available evidence suggests that, for these pathogens, therapy of arthritis with itraconazole is equivalent to that with amphotericin B. The role of voriconazole, posaconazole, and isavuconazole is unclear in the treatment of the endemic mycoses. Coccidioidal arthritis may require surgical débridement in more severe cases, but the other endemic mycoses usually respond to medical therapy alone.⁴⁰¹

For bone and joint infection caused by *Aspergillus*, aggressive surgical intervention is recommended. Voriconazole does appear to be effective for therapy, and most experts recommend it preferentially to amphotericin B preparations.^{401,442,472} Other agents with activity include posaconazole, isavuconazole, and

itraconazole, but there is little experience using these agents for *Aspergillus* arthritis. Unfortunately, although still unusual in the United States, intrinsic and acquired resistance to voriconazole and other azole antifungals may be emerging in *Aspergillus* spp.⁴⁷³ Treatment recommendations from the CDC for empirical coverage of contaminated steroid injection–related *Exserohilum* spp. osteoarticular fungal infections that do not involve the spine are for voriconazole alone for less severe infections and a lipid formulation of amphotericin B, in addition to voriconazole, for more severe infections.⁴⁷⁴ The use of combination therapy with voriconazole and caspofungin in fungal arthritis caused by *Aspergillus* has been described, but the role of this remains unclear.^{401,442}

Treatment for hyalohyphomycotic, phaeohyphomycotic, and mucormycotic osteoarticular infections, including combination therapies, is rapidly evolving.⁴⁰¹ European guidelines recommend voriconazole as preferred treatment followed by amphotericin B for bone and joint infections caused by *Fusarium* and *Scedosporium* species.⁴⁷⁵ Many experts recommend combination therapy with voriconazole plus terbinafine for *L. prolificans* (formerly *Scedosporium prolificans*) arthritis.⁴⁷⁶ For joint infections due to *Mucorales* most treatment experience is with amphotericin B. Isavuconazole has intrinsic activity against *Mucorales* and is FDA indicated for treatment of mucormycosis based on a small number of cases, but experience in joint infections due to this pathogen are extremely limited.⁴⁰¹

Mycobacterial Arthritis

Tuberculous Arthritis

Worldwide, approximately 10% to 11% of extrapulmonary tuberculosis (TB) involves the bone and joints, accounting for 1% to 3% of all TB cases.⁴⁷⁷ The incidence is higher in the developing world because of the prevalence of HIV disease. In endemic regions of the developing world, TB arthritis is mostly a disease of children and young adults, whereas in other regions, older adults and immunocompromised hosts are predominantly afflicted. Risk

factors for TB arthritis include age older than 65 years, female sex, immigration from regions with high TB endemicity, a lower socioeconomic class, incarceration, alcohol abuse, debilitating illness, IVDU, immunosuppressive drug therapy (including corticosteroids and TNF inhibitors), HIV infection, and preexisting joint disease.^{477–483} Multidrug-resistant (MDR) and extensively drug-resistant (XDR) tuberculous arthritis have been reported.^{483–485}

Clinical Features

Mycobacterium tuberculosis causes a chronic granulomatous monoarticular arthritis that is usually a result of the hematogenous dissemination associated with primary pulmonary TB.^{479,481} Articular infection may remain latent for long periods before clinical presentation. It typically involves the knee, hip, and ankle but can involve any joint.^{479,482,486–489} The clinical presentation is that of chronic arthritis, indistinguishable from that with other potential infectious or noninfectious causes (Fig. 105.7). Long-standing cases may develop an articular-cutaneous fistula, whereas TB arthritis of the wrist may involve the periarticular tendon sheaths arising with a carpal tunnel syndrome.^{486,490} Coexisting pulmonary or other extraarticular infection may be evident, but in more than half of cases it is not.^{479,482,483,486} Fever and other constitutional symptoms of TB are often absent. Interferon- γ release assay or tuberculin skin testing is positive in 85% to 90% of patients.^{479,482}

Diagnosis

To prevent unacceptable delays in diagnosis and further joint destruction, a high index of suspicion for TB arthritis must be maintained, particularly for patients who are immigrants from regions with high TB endemicity and those with other risk factors for TB. In the absence of known extraarticular TB, diagnosis requires synovial fluid analysis and, in many cases, synovial biopsy. Synovial fluid demonstrates inflammatory changes, with leukocyte counts characteristically between 10,000 and 20,000 cells/mm³, although higher counts in the range seen with bacterial arthritis may be present.⁴⁸⁶ Staining of synovial fluid demonstrates acid-fast bacilli in a minority of joint aspirates; however, mycobacterial

culture yields *M. tuberculosis* 80% of the time. The most sensitive means of confirming the diagnosis has traditionally been synovial biopsy, in which the characteristic histopathology of caseating or noncaseating granulomas can be correlated with tissue mycobacterial staining and culture, the latter being positive in about 90% of specimens.⁴⁸⁶ Mycobacterial susceptibility tests should be performed on all culture isolates.

The use of PCR assay of synovial fluid for the diagnosis of tuberculous arthritis has shown promise and can provide a rapid and minimally invasive diagnosis.^{491,492} One study found that of 98 patients with joint TB, 82.7% were positive for TB by PCR assay, 6.1% were positive by acid-fast staining, and 17.3% were positive by culture. These findings highlight the poor performance of traditional techniques when compared with PCR.⁴⁹¹ A meta-analysis of the diagnostic accuracy of the widely used, commercially available Xpert® MTB/RIF PCR assay for bone and joint tuberculosis demonstrated pooled sensitivity and specificity of 81% and 99% compared to the composite reference standard and 96% and 85% compared to culture, respectively. Results were not affected by sample type (pus vs. tissue).⁴⁹² Another potential advantage of such PCR testing is the rapid simultaneous detection of rifampin resistance, although this was not examined in this study.

FIG. 105.7 Clinical image of chronic arthritis of the knee resulting from *Mycobacterium tuberculosis*, exhibiting joint swelling, effusion, and a draining sinus fistula.

Follow for extended description

The limitations of PCR assaying include false-positive tests caused by laboratory contamination and nonavailability of cultures in some centers, particularly the developing world. Updated guidelines suggest the off-label use of FDA-cleared NAATs for TB, including PCR, on specimens collected from sites of suspected extrapulmonary TB, including synovial fluid.^{493,494} A positive NAAT result on synovial fluid confirms the diagnosis, but a negative result may not be used to exclude TB.

Imaging findings of TB synovitis consist initially of soft tissue swelling, but later in the disease process, synovial thickening and bone destruction may be seen. The use of MRI may help differentiate tuberculous arthritis from chronic noninfectious forms of joint inflammation, including RA.⁴⁹⁵ Poncet disease, a reactive arthritis associated with pulmonary or extrapulmonary tuberculosis, can also mimic actual complicating TB infection of the joint.⁴⁹⁶

Treatment

Current recommendations from the CDC for the treatment of osteoarticular TB in adults without pulmonary TB are identical to those for extrapulmonary TB (including patients with HIV or other immunosuppressive states): isoniazid (INH), rifampin (RIF), ethambutol, and pyrazinamide for 8 weeks (ethambutol may be discontinued in isolates known to be without drug resistance), followed by INH and RIF to complete 6 months of therapy (see Chapter 255).⁴⁹⁷ Directly observed therapy is advisable for most patients, and adjuvant corticosteroids are generally not recommended. If treated early, residual joint disease is uncommon and negligible. Data are scarce regarding treatment of osteoarticular MDR TB. Treatment of MDR and XDR TB relies on susceptibility testing and is changing sufficiently rapidly that guidelines are useful but rapidly become out of date (see Chapter 255).⁴⁸⁵ Adjuvant surgical therapy is required only for reconstruction of unstable joints or osseous structures or drainage of periarticular abscesses.⁴⁹⁸ Surgical therapy may also be considered in cases in which optimal therapy is not possible because of toxicity or resistance.

Nontuberculous Mycobacterial Arthritis

Nontuberculous mycobacterial arthritis, unlike that caused by *M. tuberculosis*, is usually caused by direct percutaneous inoculation of these Page 1350opportunistic mycobacteria into the skin, joint, or periarticular structures. Soil and water, and rarely air, are the usual sources of these uncommon joint infections. *Mycobacterium avium* complex, *M. kansasii*, *M. marinum*, *M. xenopi*, and *M. terrae* are the most common causes, although virtually any nontuberculous

mycobacterium may be involved.^{499–502} Skeletal and joint infection caused by the so-called rapid growers (e.g., *M. chelonae*, *M. fortuitum*, *M. abscessus*) may follow trauma, intraarticular injection, or surgery.^{500,503–506} These opportunistic pathogens are usually associated with water, including tap water, and thus may invade through large or small breaks in the skin. On occasion, nontuberculous mycobacterial joint infection is acquired through the hematogenous route, usually in immunocompromised hosts, and has been reported with *M. avium* complex, *M. haemophilum*, *M. kansasii*, *M. fortuitum*, *M. abscessus*, and *M. chelonae* (see Chapters 257 and 258).^{507–510}

Chronic monoarticular arthritis and tenosynovitis caused by nontuberculous mycobacteria are well described.^{499,500,511,512} *M. marinum* arthritis most often affects the small joints of the hand or wrist and their overlying tendon sheaths.^{501,513} Infection is acquired through preexisting or concomitant skin trauma, often minor, during contact with fresh, brackish, or salt water or with marine animals, such as fish and crustaceans. Diagnosis of nontuberculous mycobacterial joint or tendon infections usually requires biopsy of infected structures for histopathologic examination and culture. If *M. marinum* is suspected, cultures should be incubated at 30°C in addition to 37°C, to facilitate growth of this organism.⁵¹¹ Treatment varies depending on the identity of the isolated mycobacteria. Susceptibility testing can be performed in reference laboratories and is useful in directing therapy for some of these organisms. Initial therapy with a combination of three drugs—clarithromycin or azithromycin, rifampin, and minocycline—is recommended for serious joint infections pending susceptibility testing.

Mycobacterium leprae has also rarely been associated with an inflammatory arthritis that often presents in conjunction with erythema nodosa leprosum (type 2 lepra reaction) (see Chapter 256).³⁹⁷ This systemic reaction is caused by circulating immune complexes and, in its most severe form, presents with diffuse soft tissue, eye, ear, and nerve swelling with acute polyarthritis, particularly of the hands, wrists, and ankles. In addition, advanced

Hansen disease may also be associated with chronic neuropathic (Charcot) arthritis of the peripheral joints.

Septic Bursitis

Septic bursitis is common, usually affecting the subcutaneous olecranon, prepatellar, or infrapatellar bursae.^{514–516} It accounts for 1 to 12 cases per 10,000 hospitalizations and an incidence of 10 per 100,000 persons. Almost all cases are in males, particularly between the ages of 40 to 60 years. About one-fourth of patients will have at least one comorbidity, particularly diabetes.⁵¹⁶ Bacteria are most often introduced through trauma or accidental percutaneous punctures and, very rarely, through intrabursal injection of corticosteroids or hematogenous dissemination.^{514–519} Septic olecranon bursitis is four times more likely than septic prepatellar bursitis. Infection of deep bursae is rare and is usually secondary to contiguous spread from an adjacent joint infection.

More than 80% of septic bursitis is due to *S. aureus*, with the remainder due to *Streptococcus* spp. and various gram-negative bacteria, mycobacteria, and fungi.^{514,516–518,520,521} Olecranon septic bursitis caused by the algal agent *Prototheca* has been reported.^{514,522,523} Of note, infectious bursitis is more common during the summer months, especially that caused by *S. aureus*.

Patients with superficial bursitis present with painful swelling, redness, and increased warmth of the afflicted bursae (Fig. 105.8).^{514,524} Evidence of an overlying cutaneous injury or lesion is often evident. Unlike septic arthritis, range of motion in the joint is not very restricted. However, in moderate or severe cases, pain may be extreme and the range of motion of the underlying joint reduced. Soft tissue edema and erythema may extend along the extremity and circumferentially about the joint. Associated systemic symptoms potentially include fever, chills, and malaise. Noninfectious causes of bursitis, especially that involving the olecranon bursa, can occur in conjunction with several systemic conditions, such as RA, gout, pseudogout, chondrocalcinosis, and pigmented villonodular synovitis, and must be differentiated from septic bursitis.⁵²⁴ Sterile traumatic bursitis usually involves the

prepatellar bursa and presents with a paucity of systemic symptoms.

FIG. 105.8 Photograph of septic olecranon bursitis due to *Staphylococcus aureus*.

Follow for extended description

The diagnosis of septic bursitis is made by aspirating the affected bursae and analyzing the fluid for WBC count and crystals. The WBC count threshold for a presumptive diagnosis of septic bursitis is greater than 1000 cells/mm³ and is much lower than that for septic arthritis.⁵²⁴ One study found a median bursal fluid leucocyte count of 6050 cells/mm³, although in 32% of cases it was less than 2000 cells/mm³.⁵²⁵ Gram staining and culture of bursal fluid should be performed. For chronic septic bursitis, mycobacterial and fungal smear and culture are indicated. To distinguish septic olecranon bursitis from acute arthritis of the elbow, it is helpful to evaluate whether pain is worsened by elbow extension versus flexion. Bursal pressures are increased in the latter case. Thus for patients with septic bursitis, elbow pain is increased with joint flexion, whereas for patients with septic arthritis, synovial pressures are increased during elbow joint extension, and pain is greatest in this position.⁵¹⁸ Arthrocentesis should be performed if there is suspicion of joint involvement. Imaging with MRI or CT, followed by directed needle aspiration, is necessary to evaluate patients for deep bursal infection.

Treatment for septic bursitis includes antibiotics and daily aspiration of the bursae until sterile fluid is obtained.^{515,525–527} In some acute cases bursectomy may be required. Endoscopic and open bursectomy have similar outcomes with the former having shorter hospital stay and less postoperative pain.⁵²⁸ Antimicrobial selection should be based on bursal aspirate Gram stain as outlined in Table 105.5. Oral antimicrobials with antistaphylococcal activity are initially indicated for mild cases afflicting healthy patients with good access to medical care. Because of the high prevalence of CA-MRSA in many localities, clindamycin, TMP-SMX, minocycline, and linezolid are appropriate choices because of their reliable activity

against this pathogen. For more severe cases or patients with chronic illness or immunosuppression, IV antimicrobials should be selected. In both cases definitive antibiotic therapy should be selected based on the identity and susceptibilities of cultured bacteria and continued to complete a 14- to 21-day course. The prognosis of prepatellar or olecranon septic bursitis is generally quite good, although recurrences are common and may require bursectomy when infection is quiescent. Page 1351

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